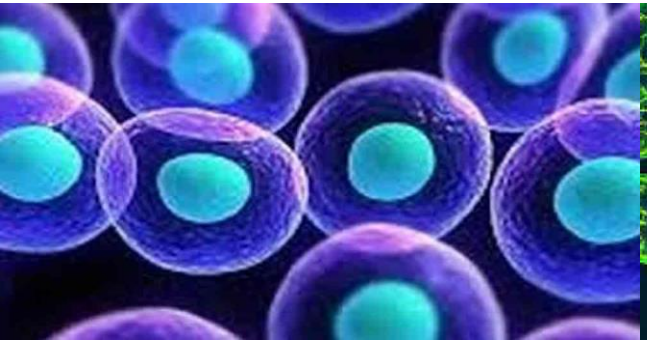
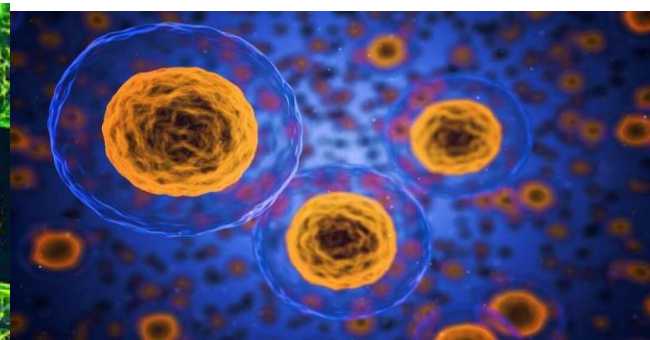
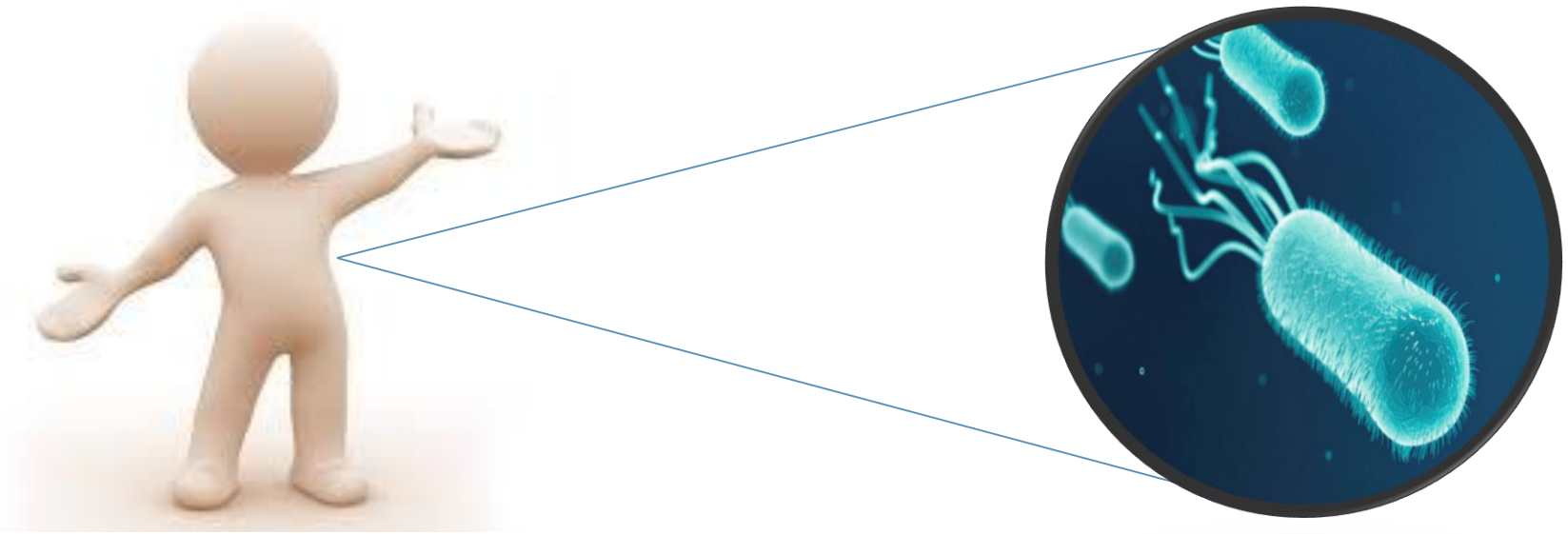
Part 11

# Prokaryotic cells





LOOK AT YOU



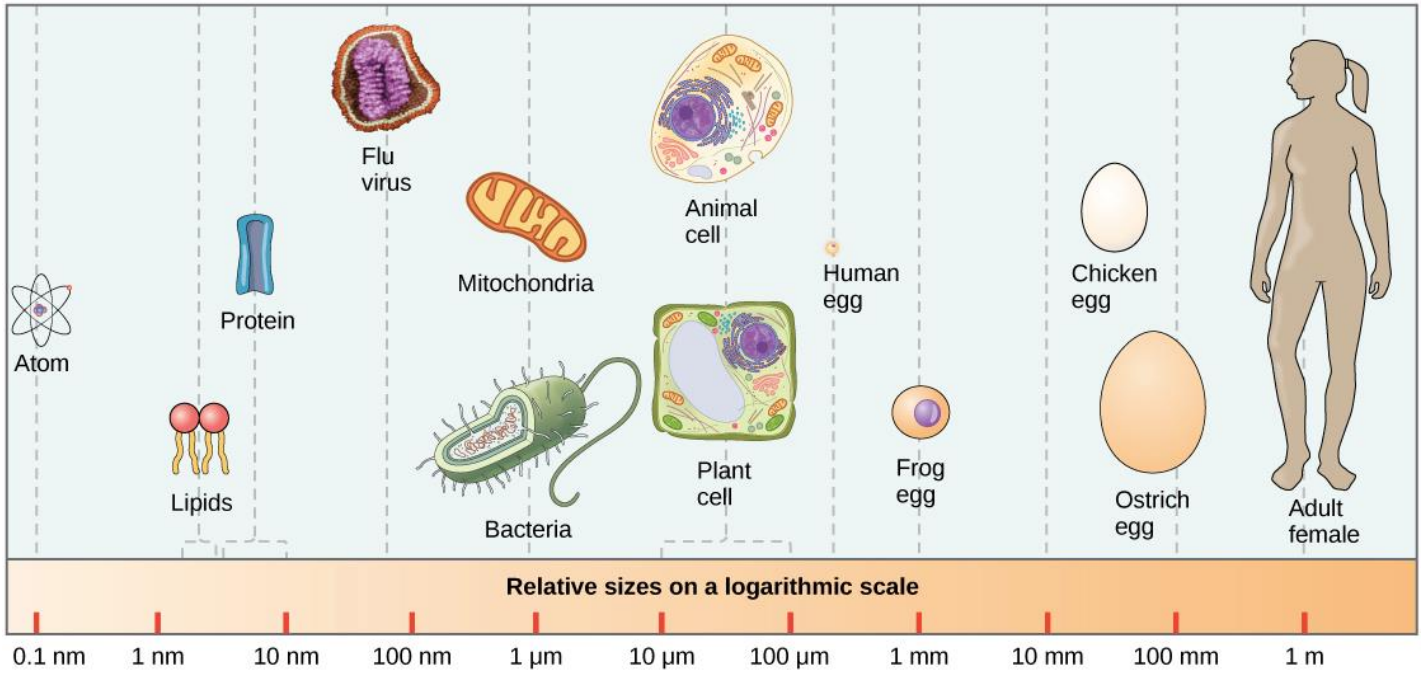
YOU ARE NOT ALONE

You are home to ~ **100 trillion** bacterial cells

Number of bacteria cells inhabiting your body= (your cells) x 10-100 times

1%–3% of the **body mass** is composed of **bacteria**

# Cell size

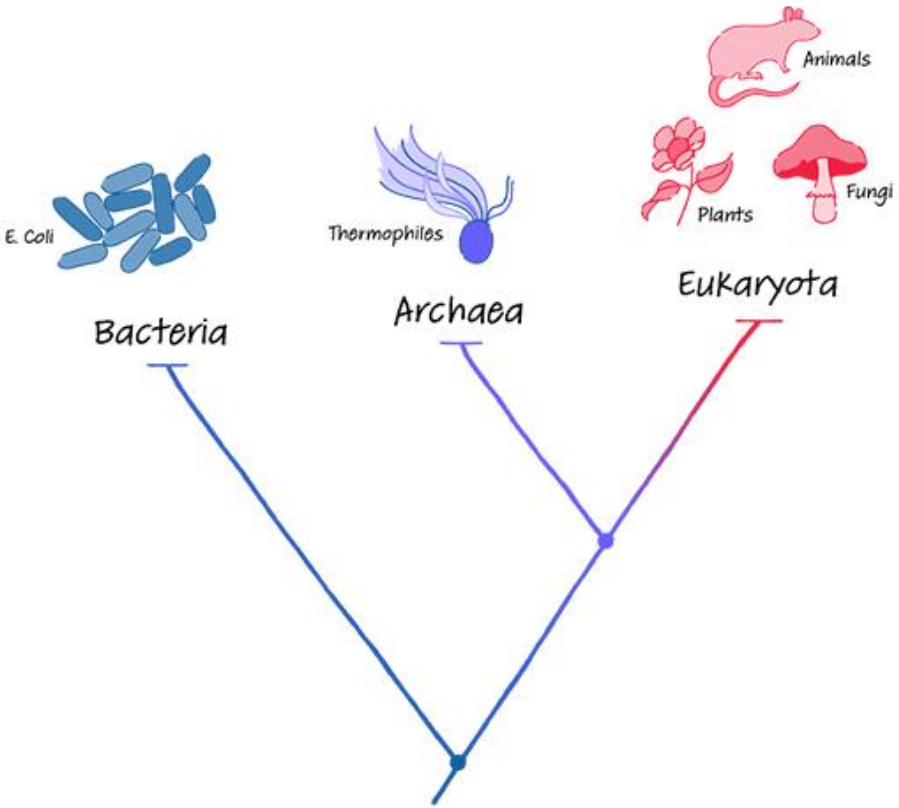



Prokaryotic cells : 0.1 ~to 5.0 µm in diameter

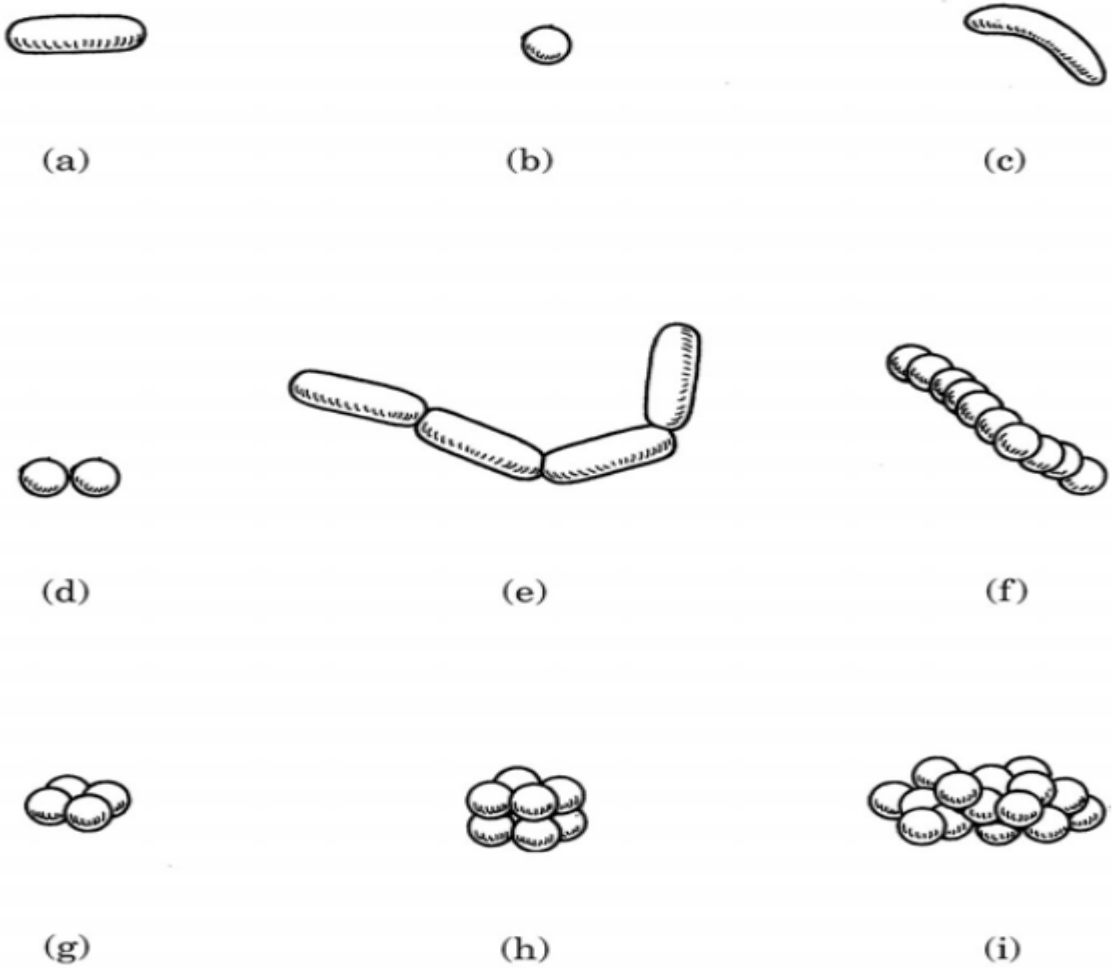
Eukaryotic cells: 10-100 µm in diameter

Why prokaryotic cells are so small?

# Classification



# Bacterial diversity: Shape



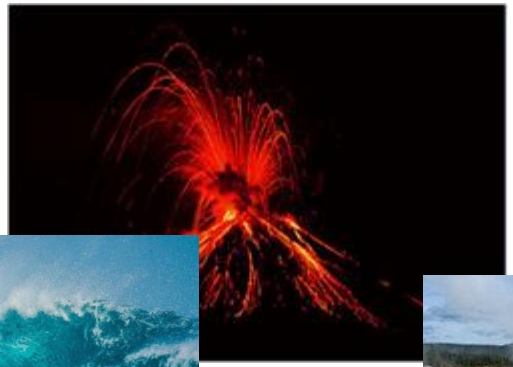

Bacterial shapes. Most bacteria are (a) rod shaped, (b) spherical or (c) curved. These basic shapes may join to form (d) pairs, (e and f) chains, (g) sheets, (h) packets or (i) irregular aggregates

# Bacterial diversity: habitant

Bacteria live in all ecosystems

- On/in plants & animals (living/dead)
- In the soil/water/air
- In extreme environment (hot, salt, acid, base…)





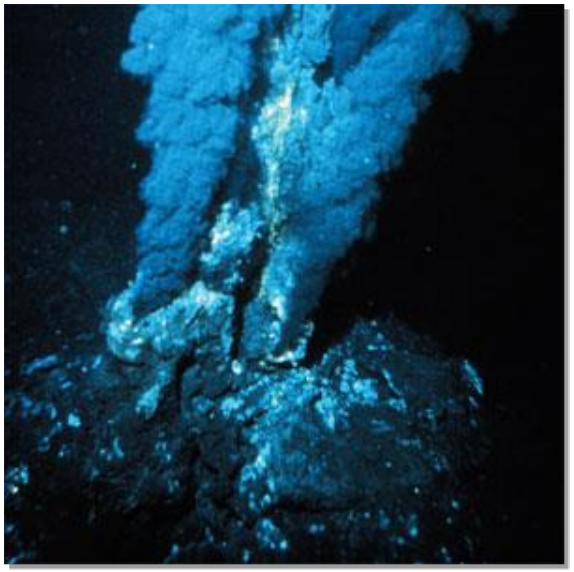




# Bacterial diversity: metabolism

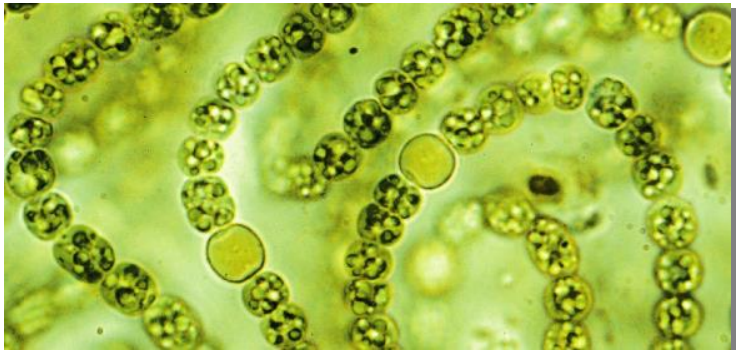
Carbon source
- **Autotrophs**
- **Heterotrophs**

Energy source
- **Phototrophs** (use the light)
- **Chemotrophs** (use the chemical)



Oxygen require
- **Aerobic** (oxygen-requiring)
- **Anaerobic** (non-oxygen-based) metabolism

Others: metabolize nitrogen- or sulfur-containing compounds





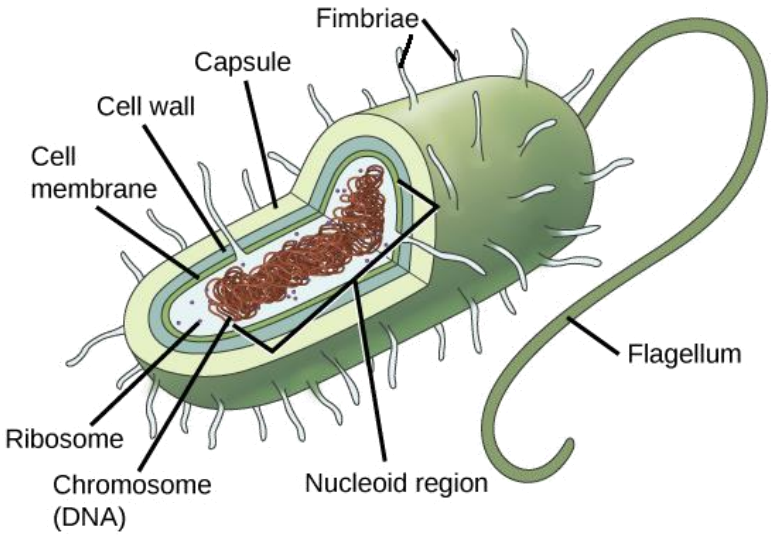
# Structure

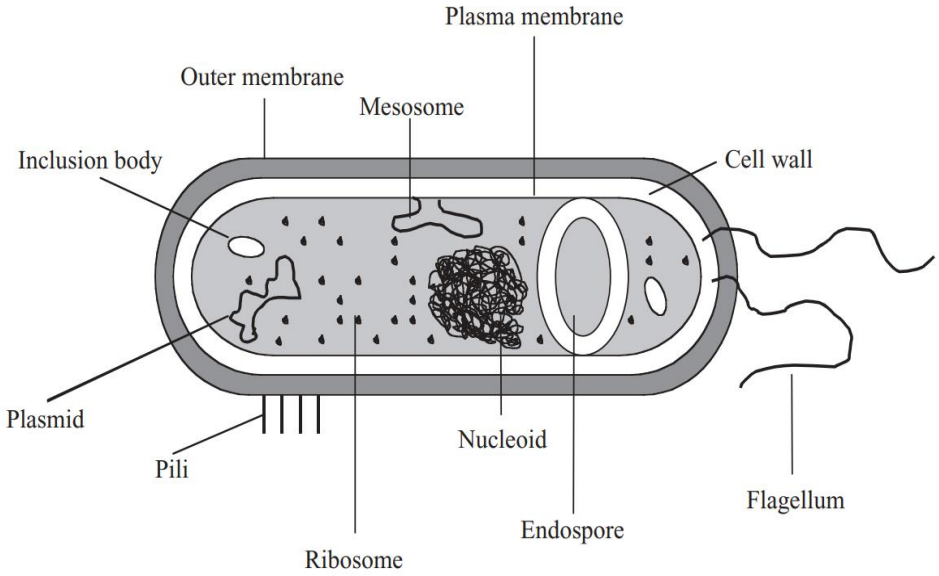
All cells share four key components:

1. Plasma membrane
2. Cytoplasm
3. DNA
4. Ribosome

Other components:

Cell wall, Capsule, Plasmid…



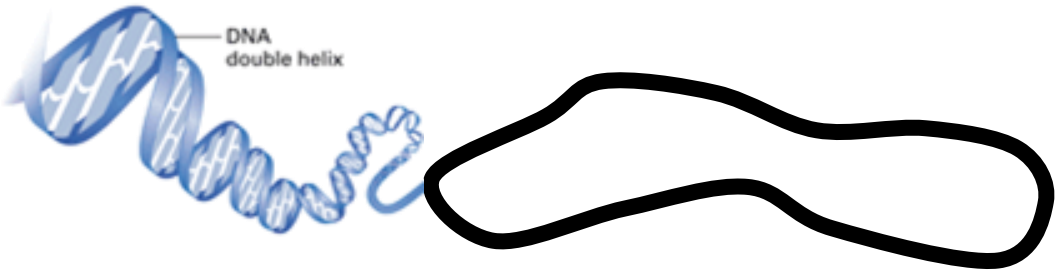
Structure

# Prokaryote vs. Eukaryote Chromosome

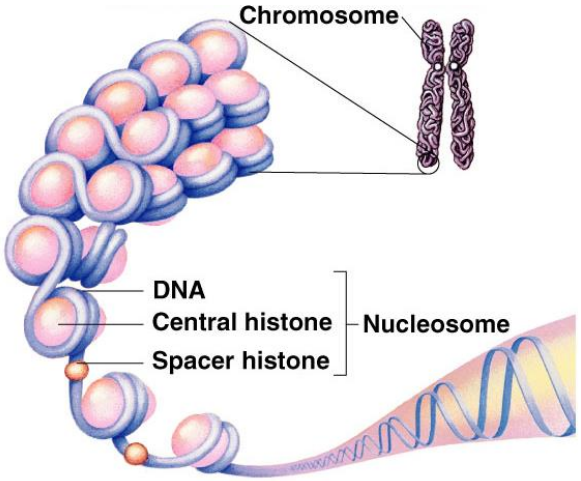


**double helix**

**Prokaryote**



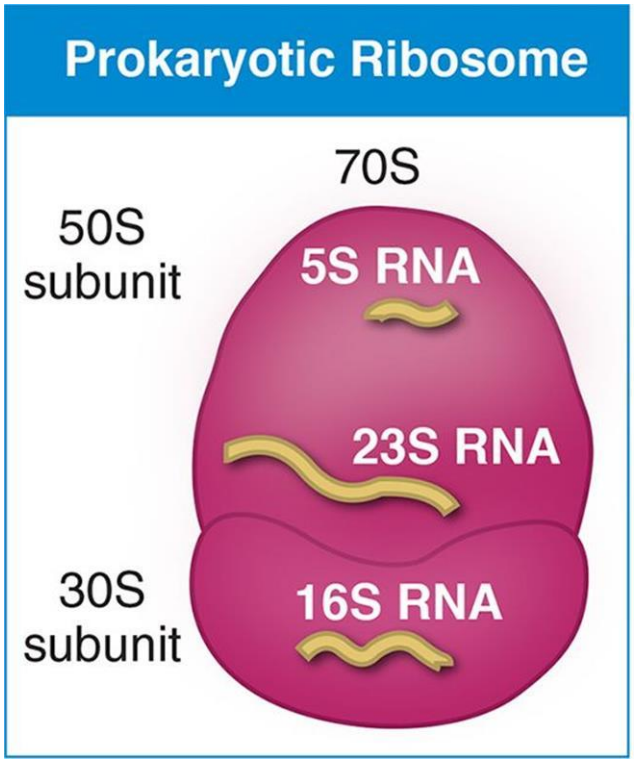

**Eukaryote**

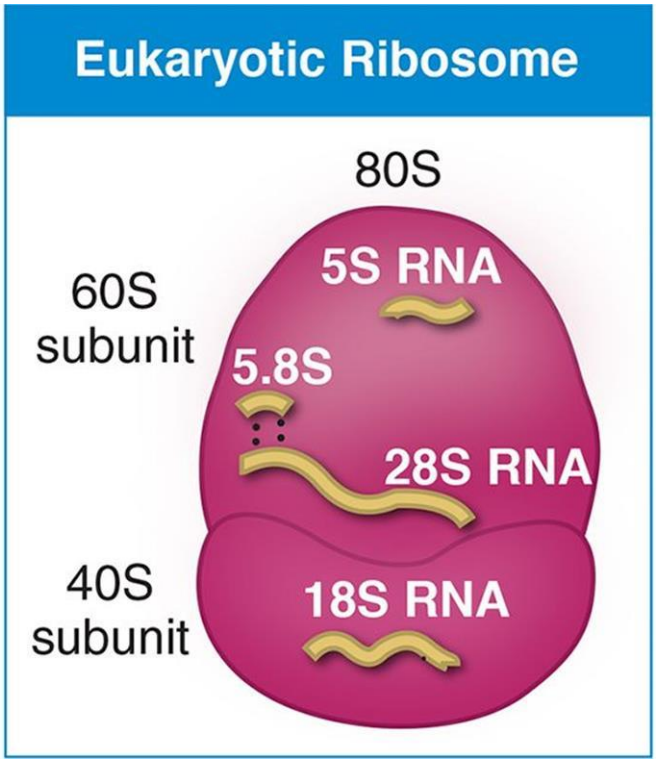



Circular chromosome, naked DNA
(not wrapped around proteins)

Structure

# Ribosome

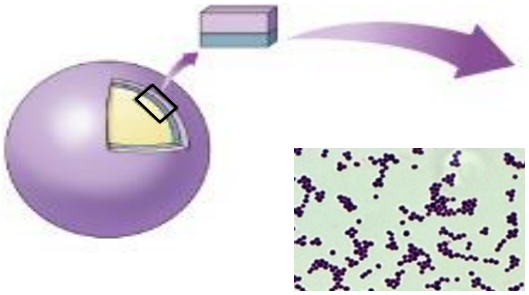


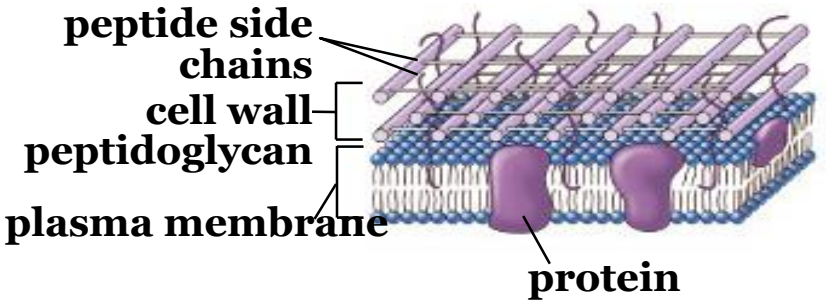



Structure

# Cell wall

## Gram-positive bacteria

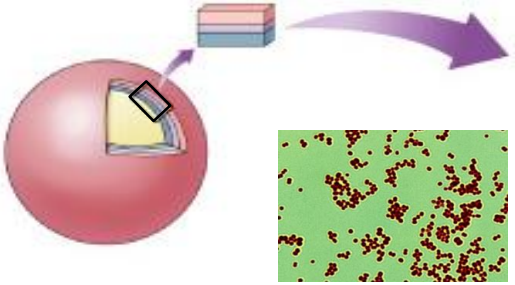


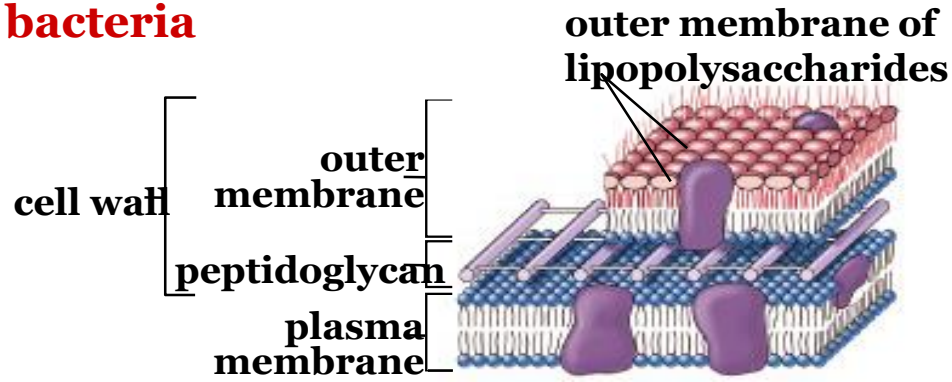



peptidoglycan = polysaccharides + amino acid chains

lipopolysaccharides = lipids + polysaccharides

## Gram-negative bacteria





# Cell wall structure

Difference between Gram-Positive and Gram-Negative Bacteria

| | Gram-Positive | Gram-Negative |
|---|---|---|
| Cell Wall | A single-layered, 20 to 80 nm | A double-layered, 8 to 10 nm |
| Peptidoglycan Layer | It is a thick layer/ also can be multilayered | It is a thin layer/ often single-layered. |
| Outer membrane | Absent | Present |
| Lipopolysaccharide | Absent | Present |
| Resistance to Antibiotic | More susceptible | More resistant |
| Gram Staining | purple-coloured | pink-coloured |
| Examples | *Staphylococcus, Streptococcus* (pneumonia) | *Escherichia, Salmonella* |

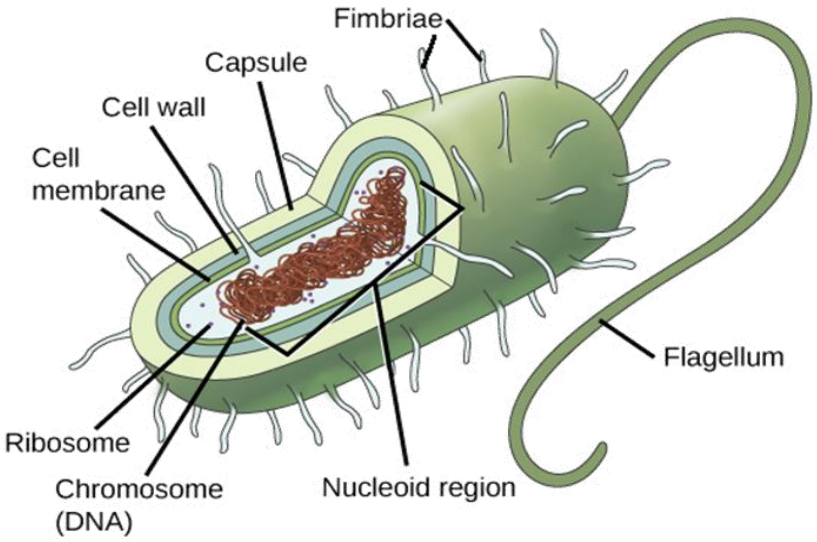
Structure

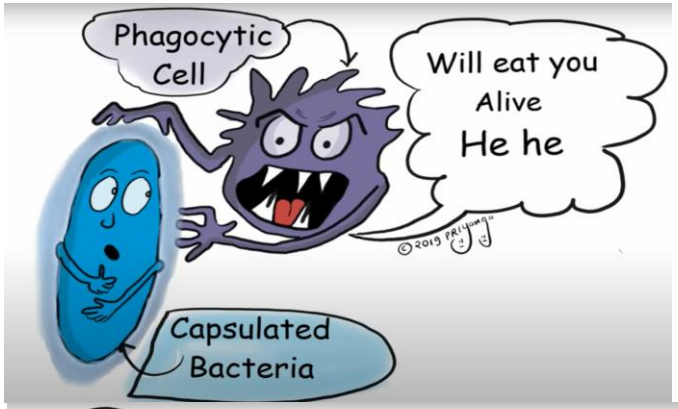
# Capsule

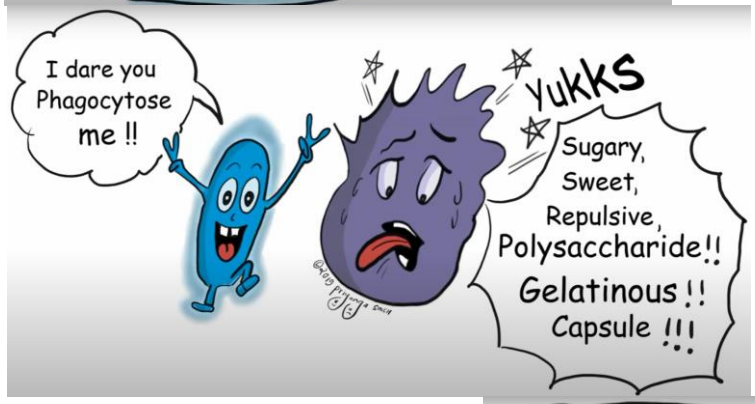
Sticky and helps the cell attach to surfaces in its environment

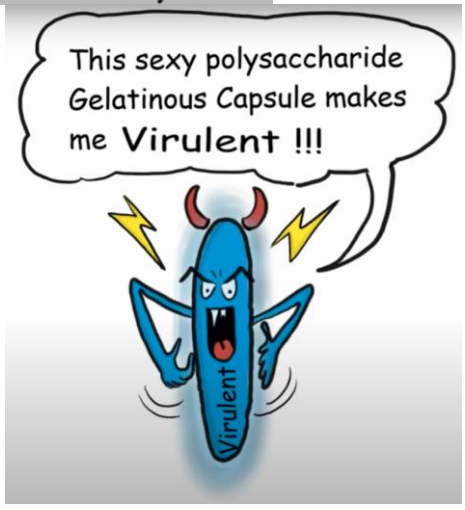
Smooth surface and negative charge of capsule prevents the phagocytosis→ virulence factor

Contain water which protects the bacteria against desiccation



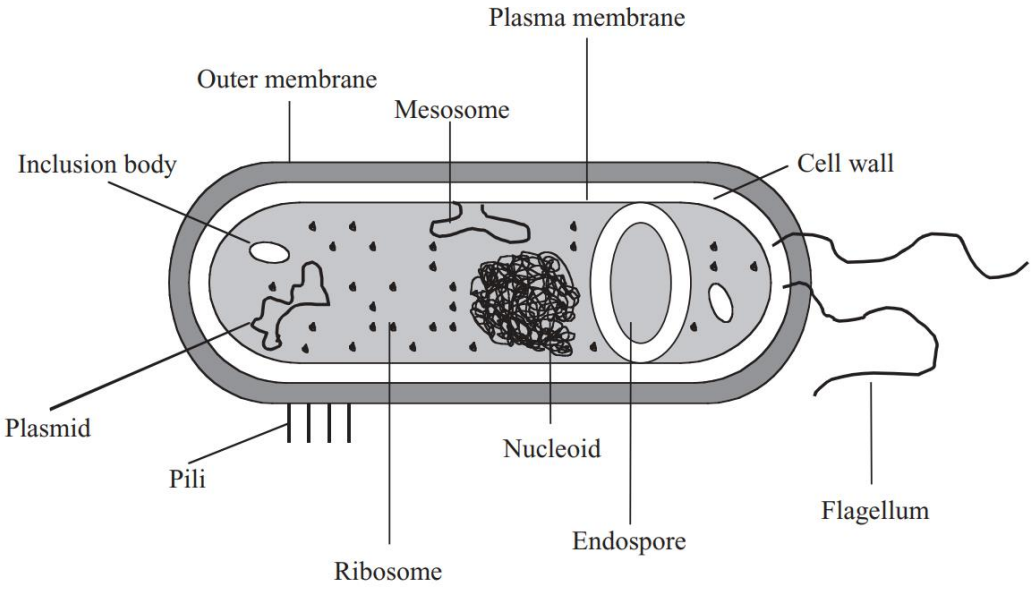



Structure

# Plasmid:

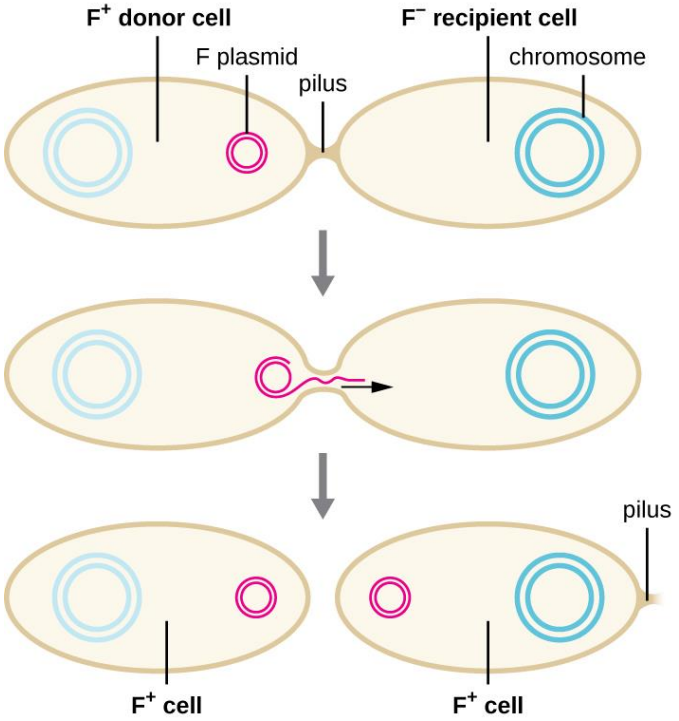
- Circular rings of naked DNA
- Extra genetic material
- Can be passed from one cell to another cell
- Used as a vectors in genetic engineering




- R plasmids: resistance plasmids
- Sex factor plasmids (F)
- Virulence plasmids: virulence factors
- Heavy metal ion resistant plasmids
- Plasmids of catabolic activity

Structure

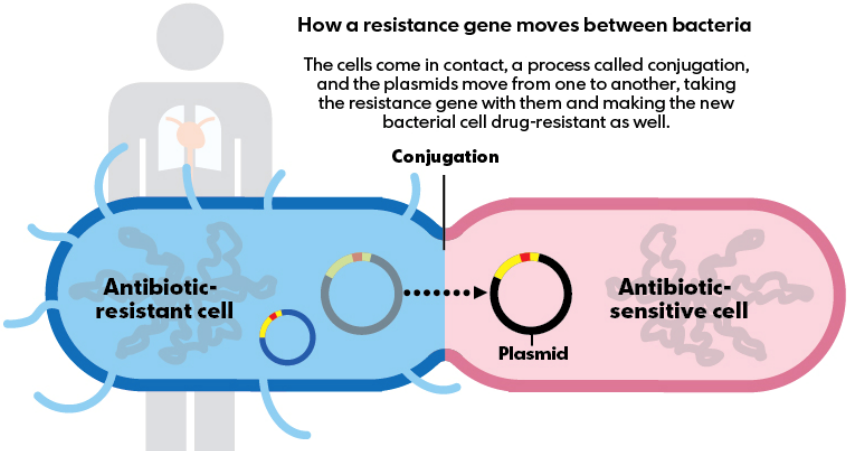
# Plasmid:




1. Pilus of donor cell attaches to recipient cell. Pilus contracts, drawing cells together to make contact with one another.

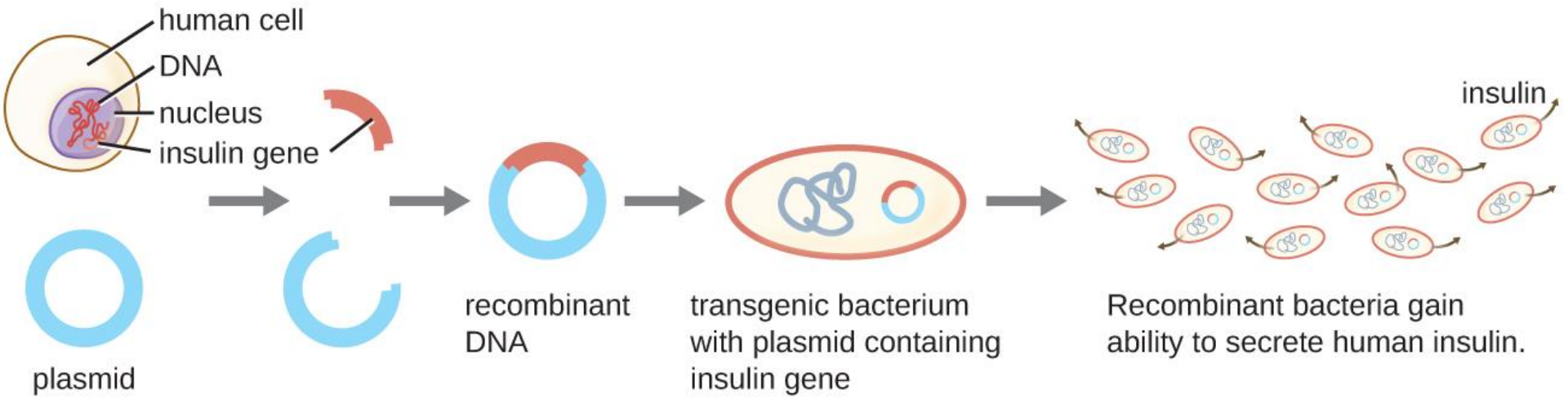
2. One strand of F plasmid DNA transfers from donor cell to recipient cell.

3. Donor synthesizes complementary strand to restore plasmid. Recipient synthesizes complementary strand to become $F^+$ cell with pilus.



Structure

Plasmid:

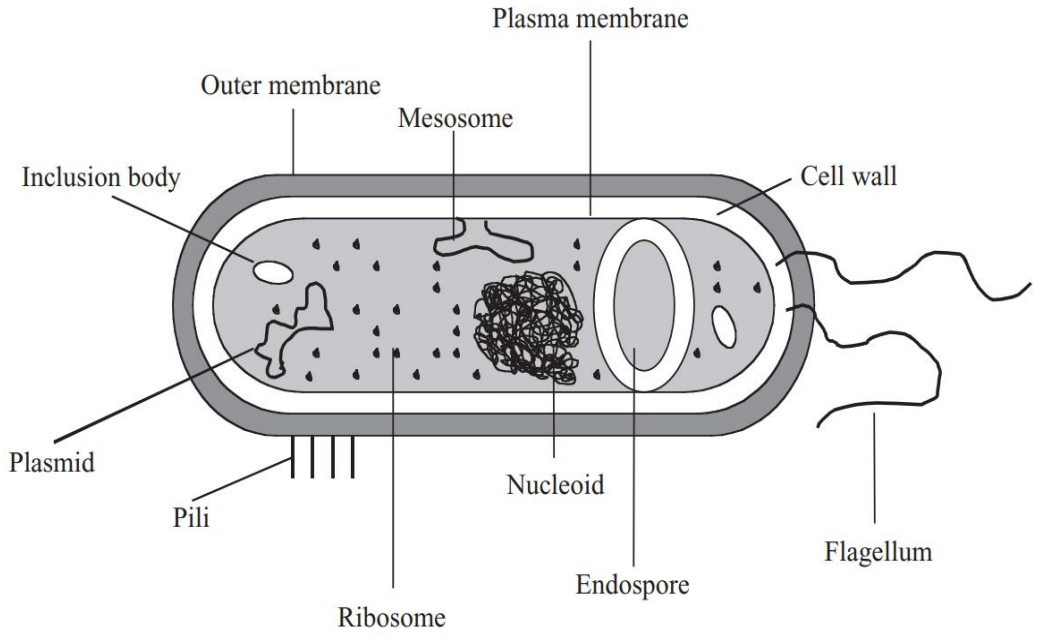



Recombinant DNA technology is the artificial recombination of DNA from two organisms.

Structure

## Mesosome:

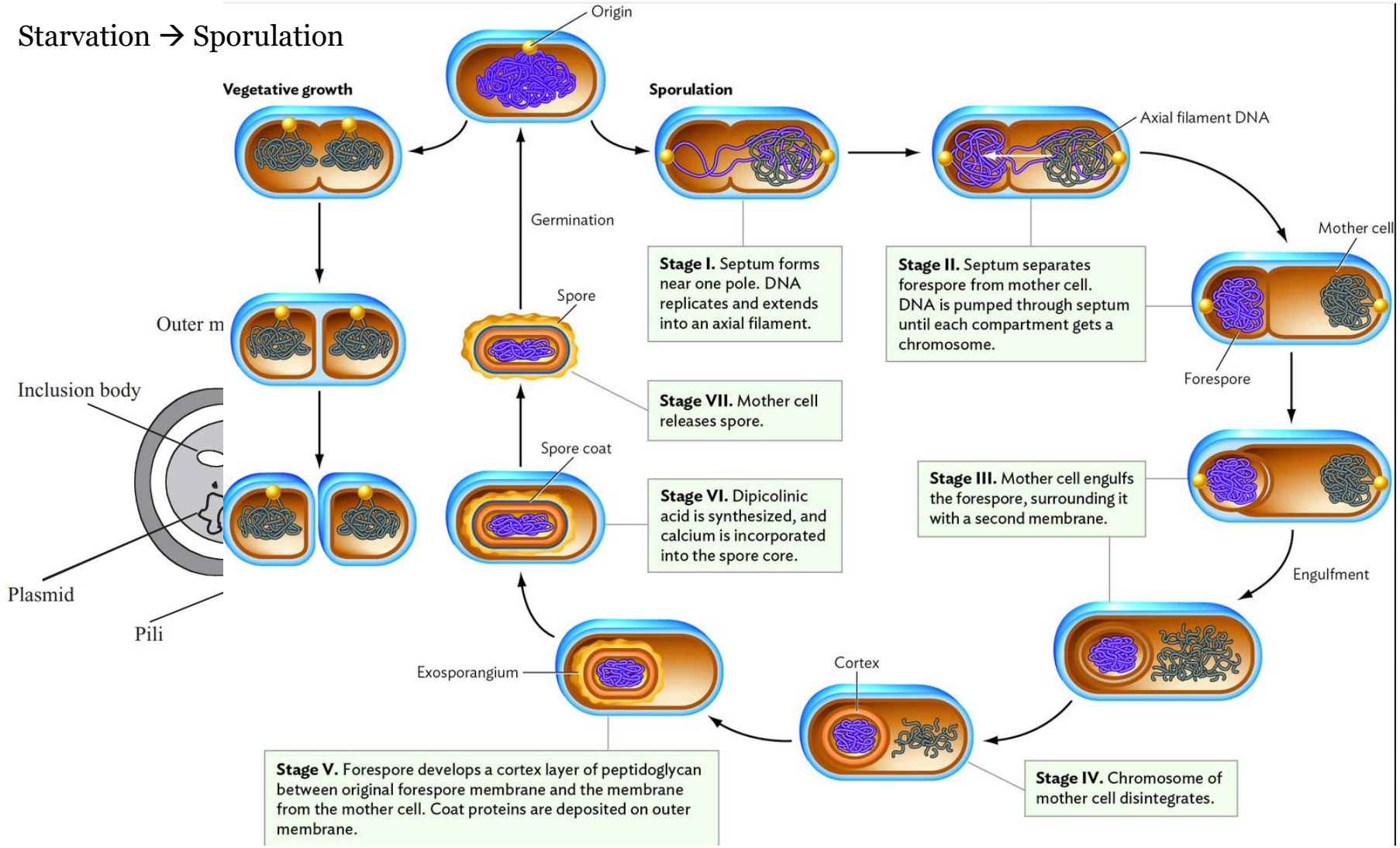
- In folding of the plasma membrane
- Plays a role in cellular respiration and movement of DNA



Structure

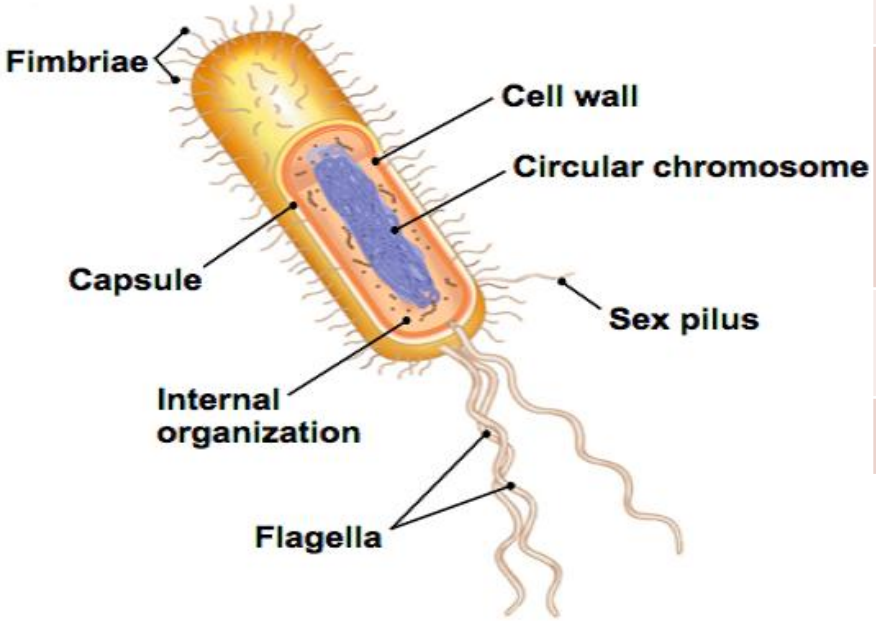
# Endospore

Starvation → Sporulation



Structure

# Fimbriae, Pilus, and Flagellum

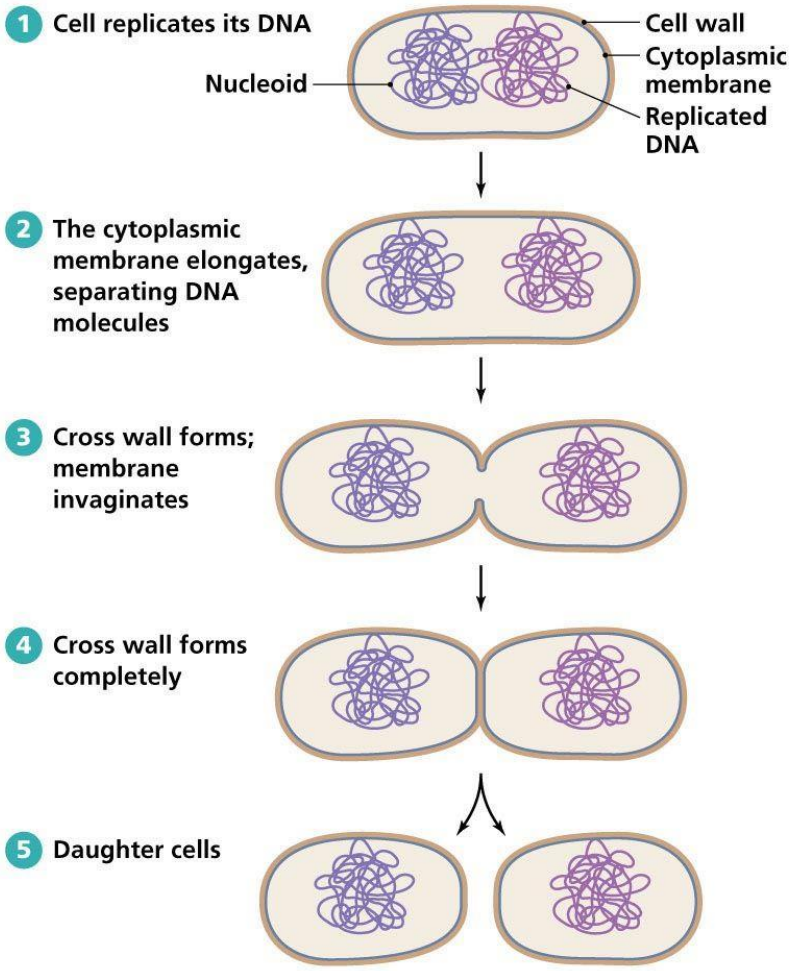



| Fimbriae | Pili |
|---|---|
| Shorter, thin, 200-400 per cell | Longer, thicker, 1-10 per cell |
| Is governed by bacterial genes in the nucleoid region. | Is governed by plasmid genes. |
| Responsible for cell to surface attachment, enable the cell to adhere the surfaces of other bacteria. | Responsible for bacterial conjugation. |
| No receptors of other. | Serve as receptor for certain viruses. |
| | |

Flagellum: For movement purpose

# Reproduction in bacteria

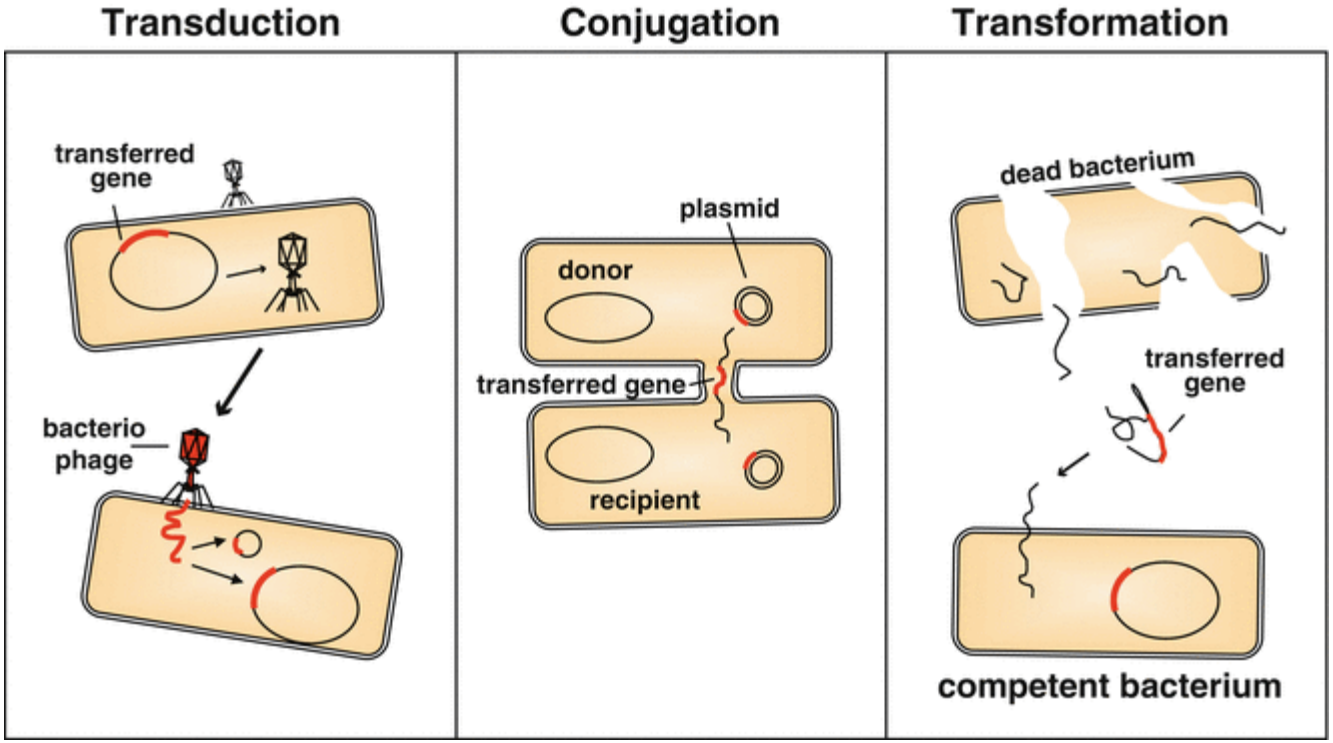
Asexual: binary fission




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# Reproduction in bacteria

Sexual:




1. Transduction: a virus transfers genetic material from one bacterium to another
2. Conjugation: use of a hollow tube called a conjugation pilus to transfer genes between cells
3. Transformation: naked DNA is taken up from the environment

# Bacteria culture

Culture on an agar plate or in a liquid culture → could be studied, analyzed

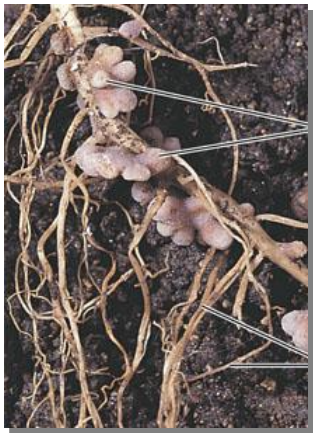
Fact: ~99% of bacteria and archaea are uncultivable

→ They are mystery





# Role of prokaryotes



Decomposers
recycling of nutrients from dead to living

Nitrogen fixation
only organisms that can fix N from atmosphere

Help in digestion (E. *coli*)
produce vitamin K & B12

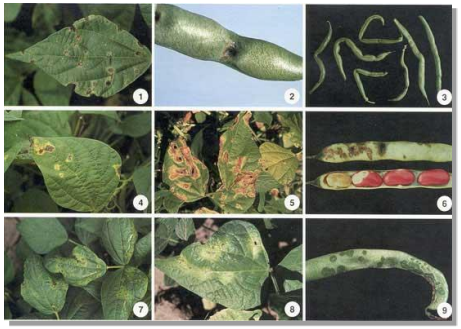



Produce foods & medicines

---

Bacteria as pathogen

Plants and animals diseases



# Antibiotic

Antibiotic is from antibiosis, meaning against life

Substances derived from a microorganism or produced synthetically to kill or suppress the growth of other microorganism



# Antibiotic

Scottish bacteriologist Alexander Fleming accidentally discovered the antibiotic in 1928

A green mold called *Pennicilium notatum* had contaminated Petri dishes were killing some of the bacteria

→ penicillin



Alexander Fleming
(6 August 1881 – 11 March 1955)

## Classification of antibiotics

1. On the basic of mechanism of action
2. On the basis of spectrum of activity
3. On the basis of mode of action

## Mechanism of action

1. Inhibition of cell wall synthesis:

   Penicillins, Cephalosporins, Bacitracin & Vancomycin

2. Inhibition of functions of cellular membrane

   Polymyxins

3. Inhibition of protein synthesis

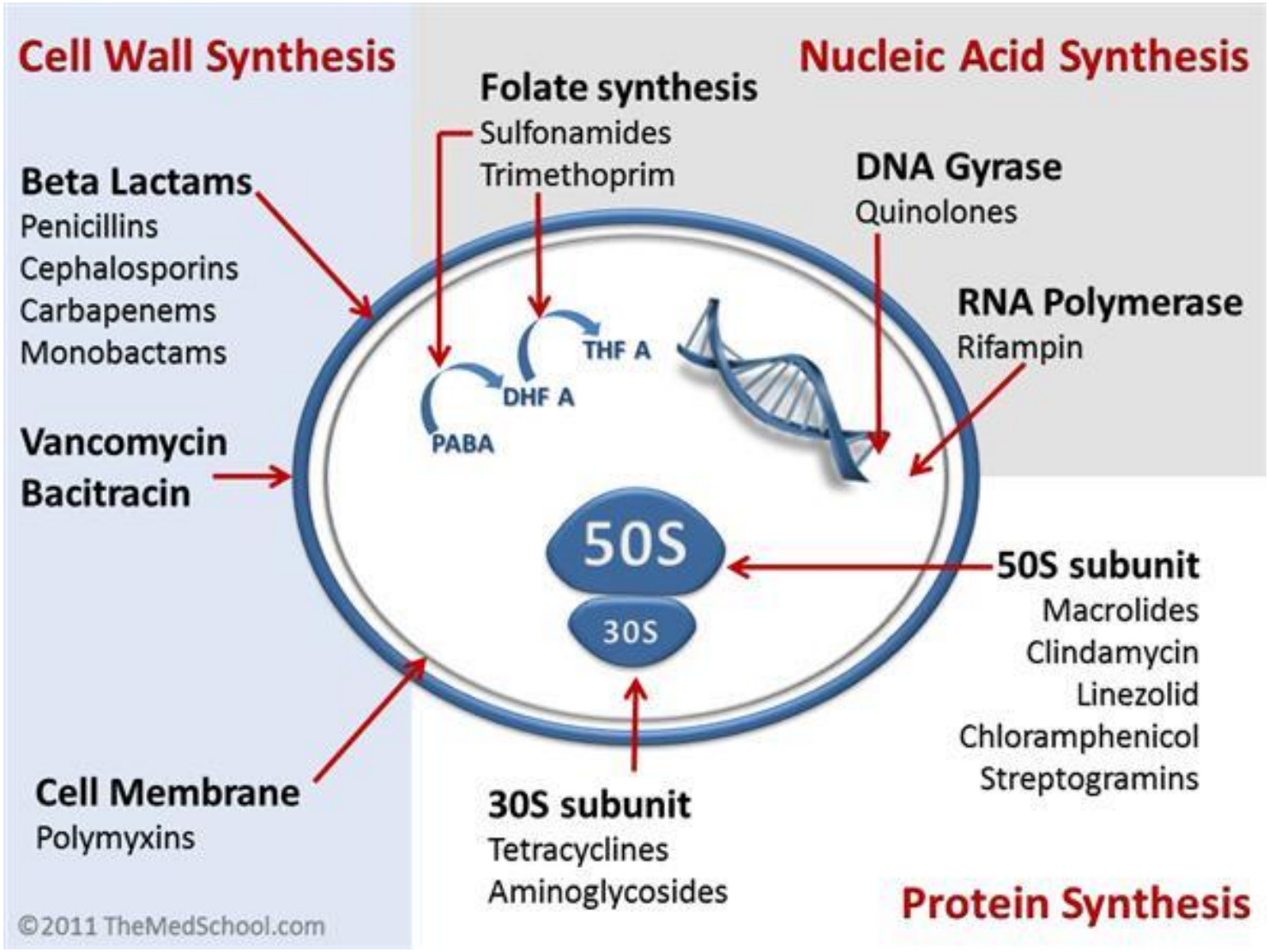
   Chloramphenicol, Macrolides & Clindamycin
   Tetracyclines & Aminoglycosides

4. Inhibition of nucleic acid synthesis:

   Quinolones, Rifampin

5. Inhibition of folic acid synthesis:

   Sulfonamides & trimethoprim

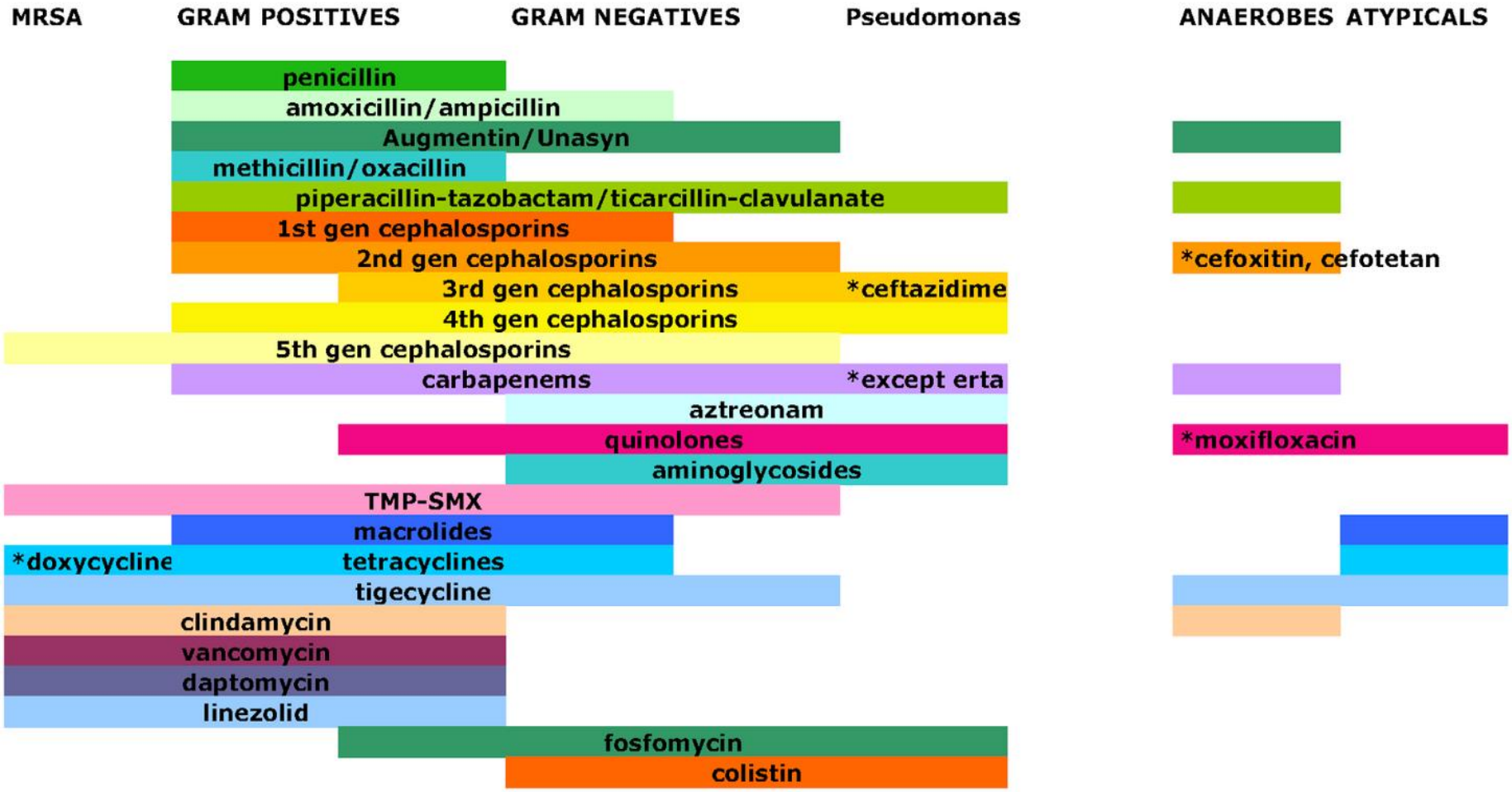

Cell Wall Synthesis
Nucleic Acid Synthesis
Folate synthesis
Sulfonamides
Trimethoprim
Beta Lactams
Penicillins
Cephalosporins
Carbapenems
Monobactams
DNA Gyrase
Quinolones
RNA Polymerase
Rifampin
THF A
DHF A
PABA
Vancomycin
Bacitracin
50S
30S
50S subunit
Macrolides
Clindamycin
Linezolid
Chloramphenicol
Streptogramins
Cell Membrane
Polymyxins
30S subunit
Tetracyclines
Aminoglycosides
Protein Synthesis
©2011 TheMedSchool.com

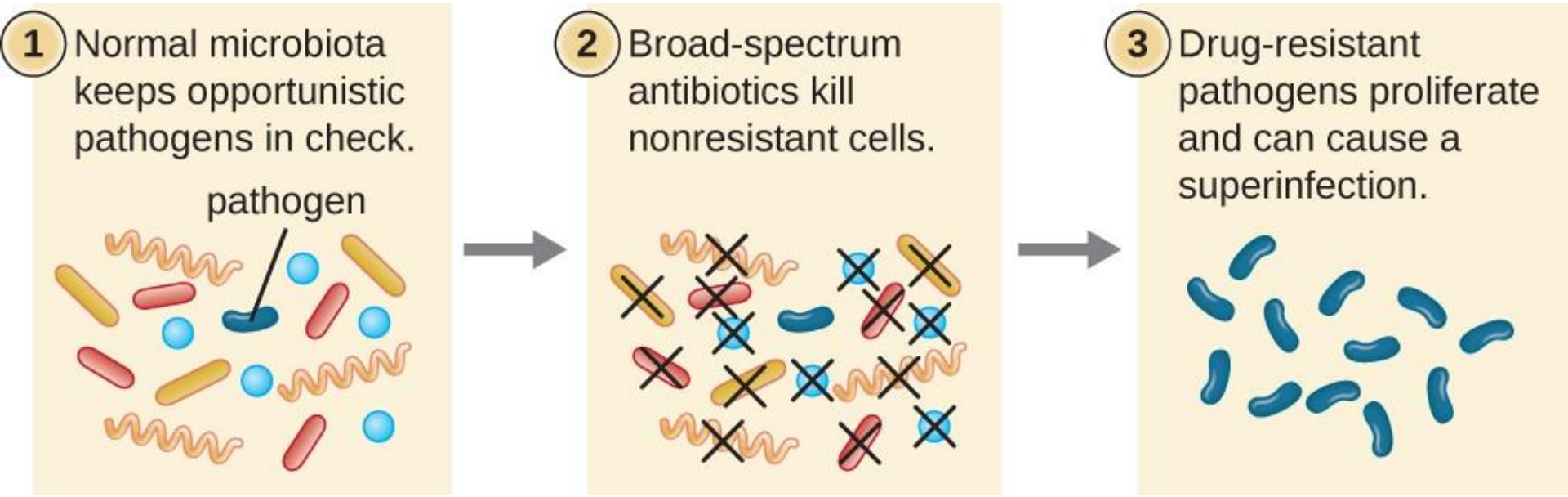
## Antimicrobial spectrum

Antimicrobial spectrum: the scope that a drug kills or suppresses the growth of microorganisms

**Narrow-spectrum**: The drugs that only act on one kind or one strain of bacteria

**Broad-spectrum:** The drugs that have a wide antimicrobial scope.(Tetracycline & Chloramphenicol)


MRSA
GRAM POSITIVES
GRAM NEGATIVES
Pseudomonas
ANAEROBES ATYPICALS
penicillin
amoxicillin/ampicillin
Augmentin/Unasyn
methicillin/oxacillin
piperacillin-tazobactam/ticarcillin-clavulanate
1st gen cephalosporins
2nd gen cephalosporins
*cefoxitin, cefotetan
3rd gen cephalosporins
*ceftazidime
4th gen cephalosporins
5th gen cephalosporins
carbapenems
*except erta
aztreonam
quinolones
*moxifloxacin
aminoglycosides
TMP-SMX
macrolides
*doxycycline
tetracyclines
tigecycline
clindamycin
vancomycin
daptomycin
linezolid
fosfomycin
colistin



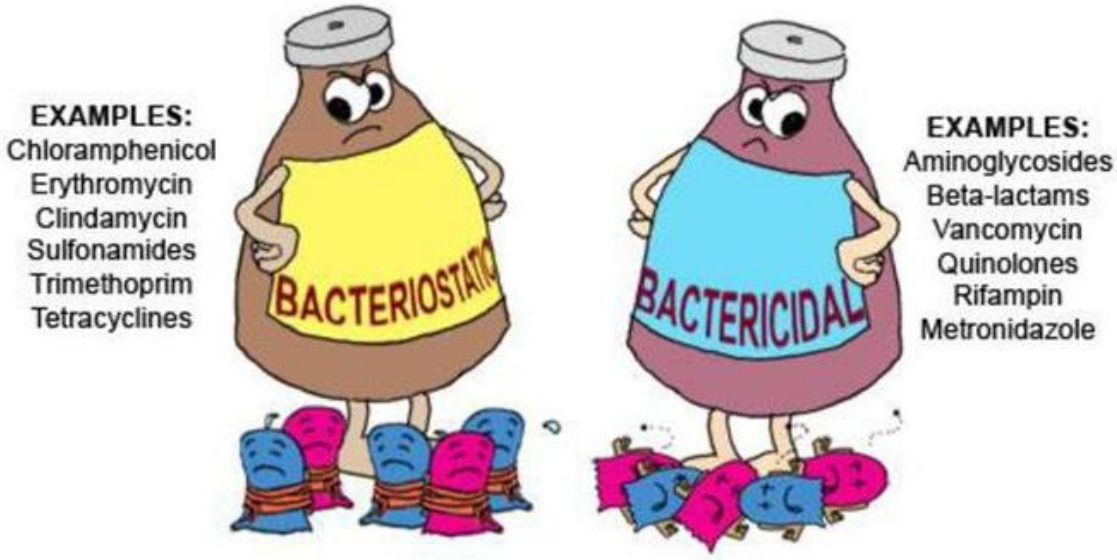

Broad-spectrum antimicrobial use may lead to the development of a superinfection

(credit: modification of work by Centers for Disease Control and Prevention)

## On the basis of mode of action

Bactericidal: antibiotics will function destroy or kill bacteria

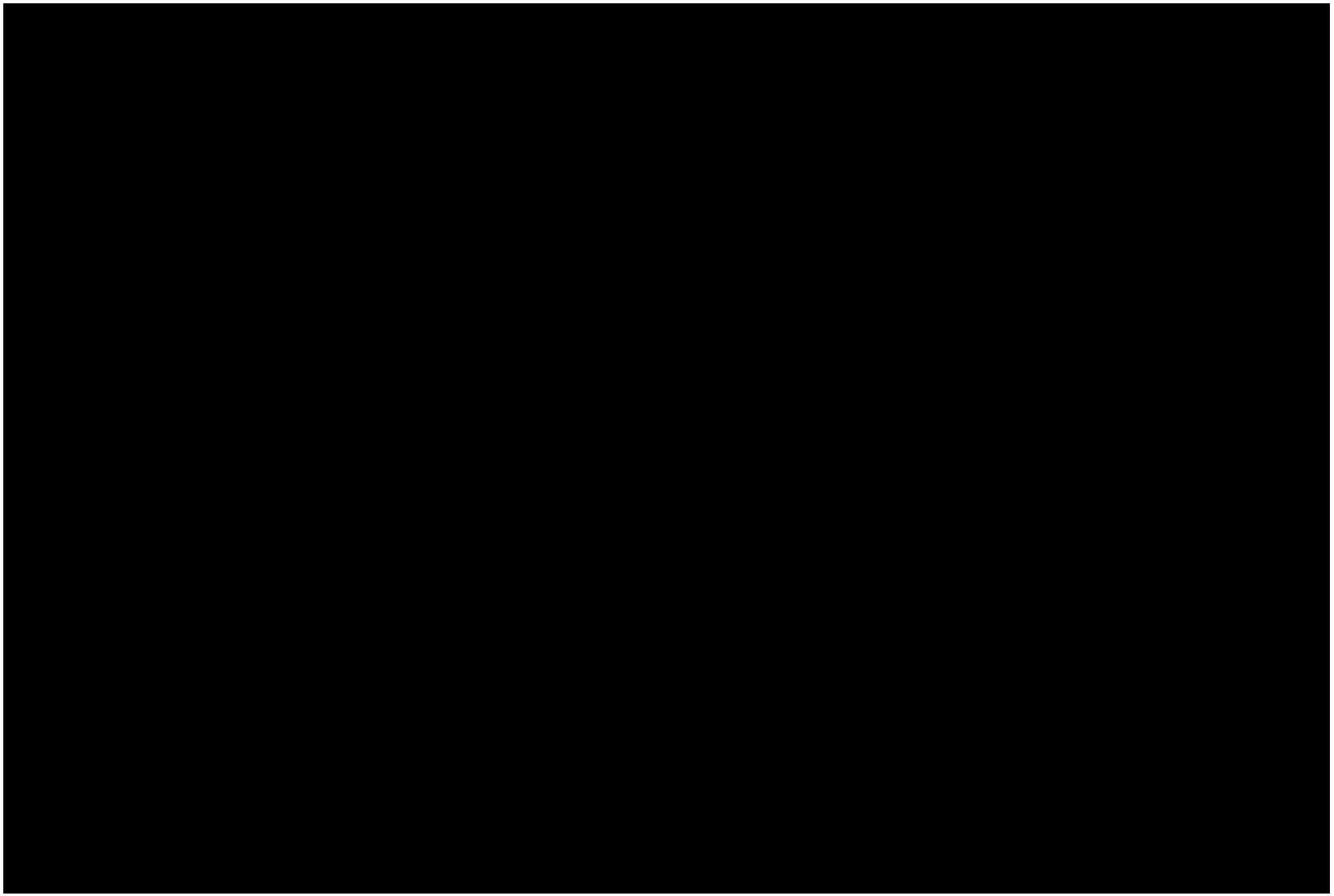
Bacteriostatic: antibiotics will function to stop or slow the growth of bacteria



Some people taking antibiotics will experience common side effects such as:



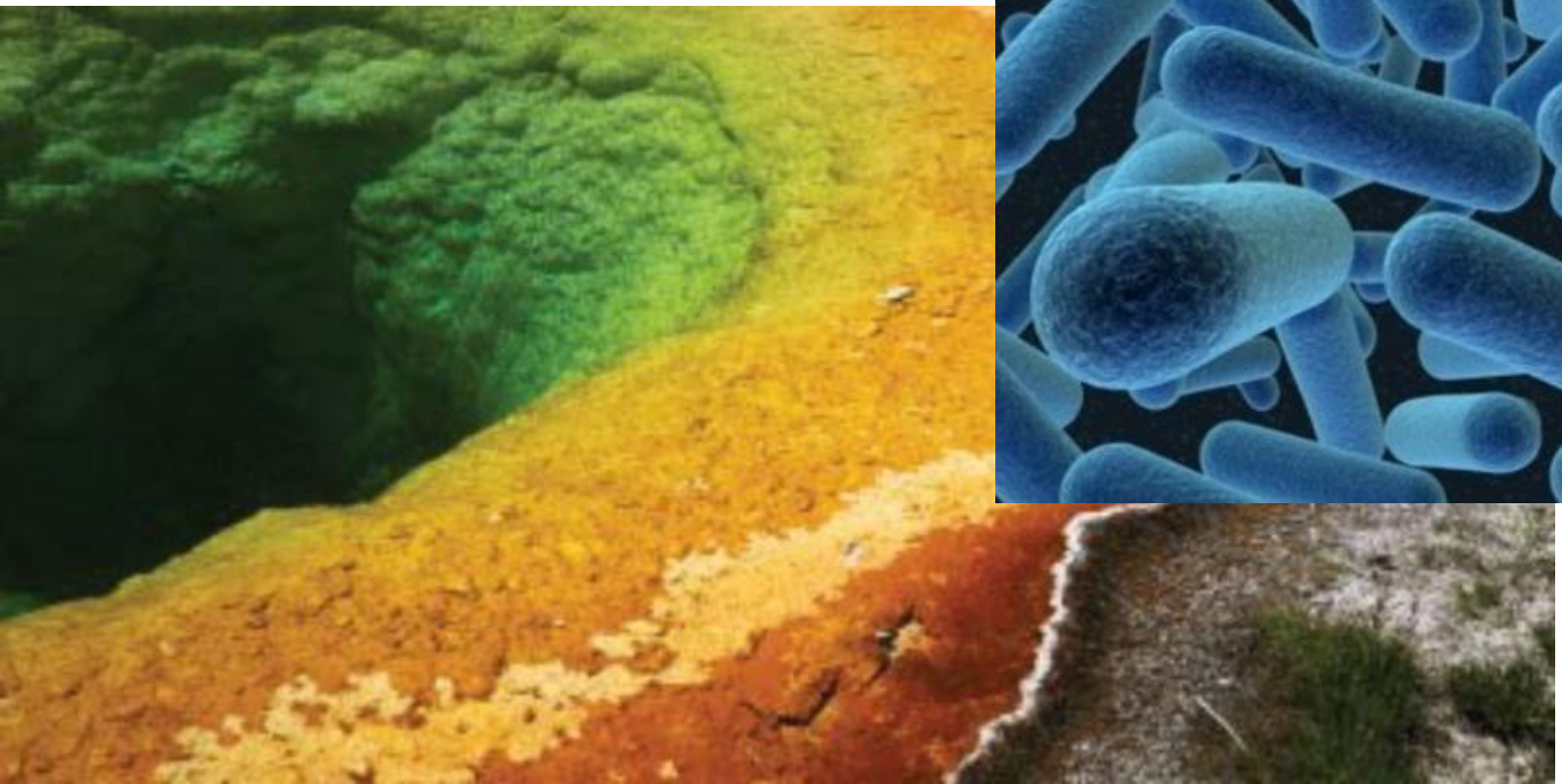
# Bacteria

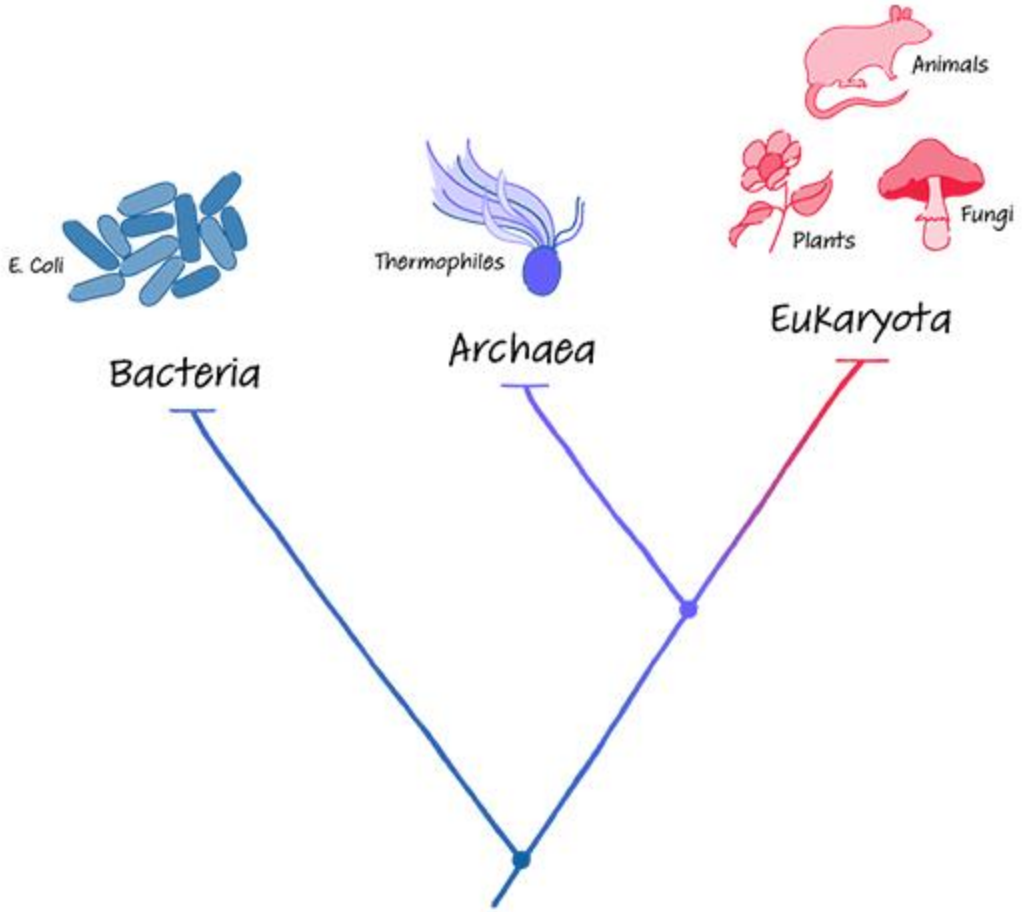


# Part 2.

# ARCHAEA

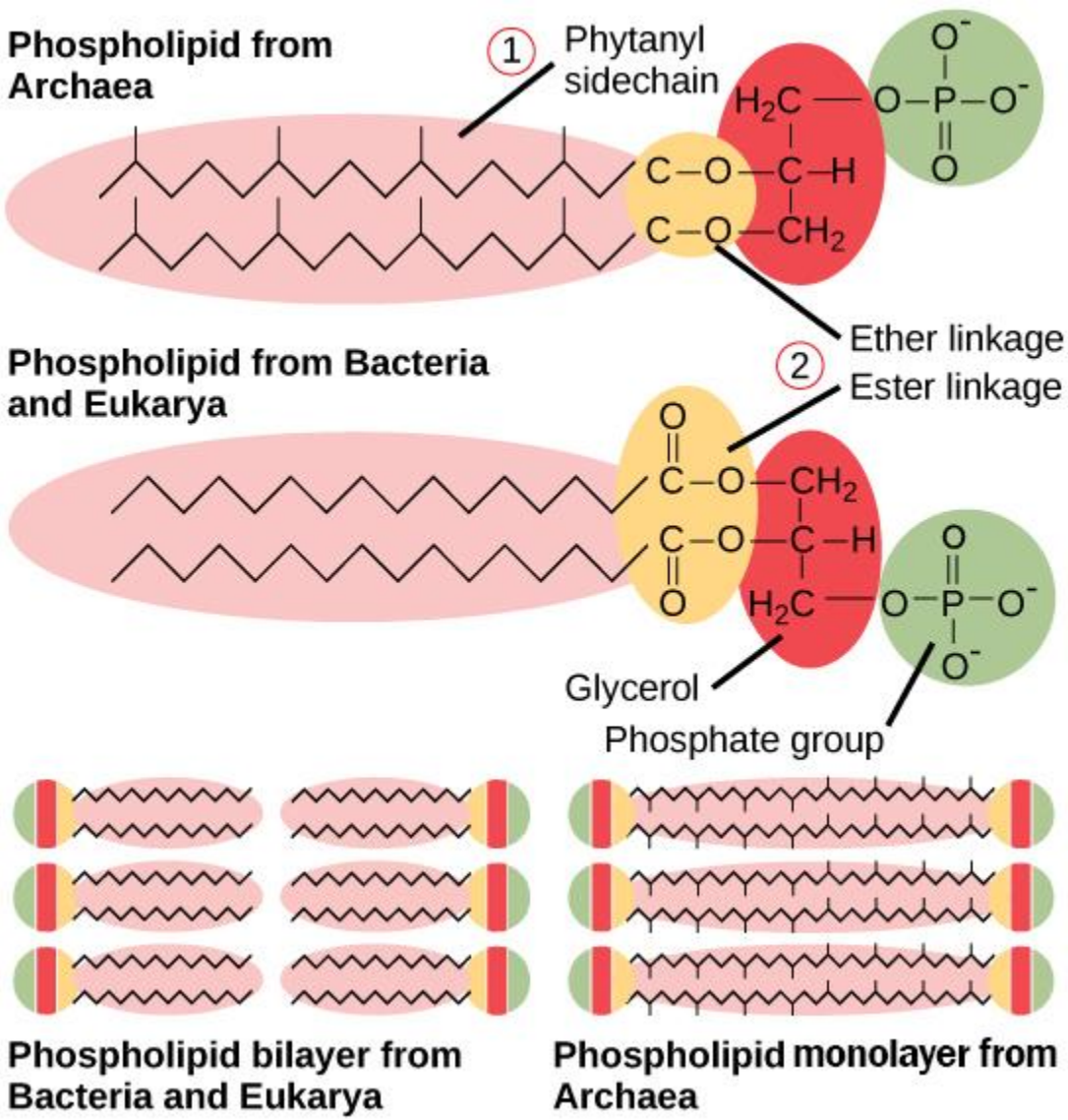




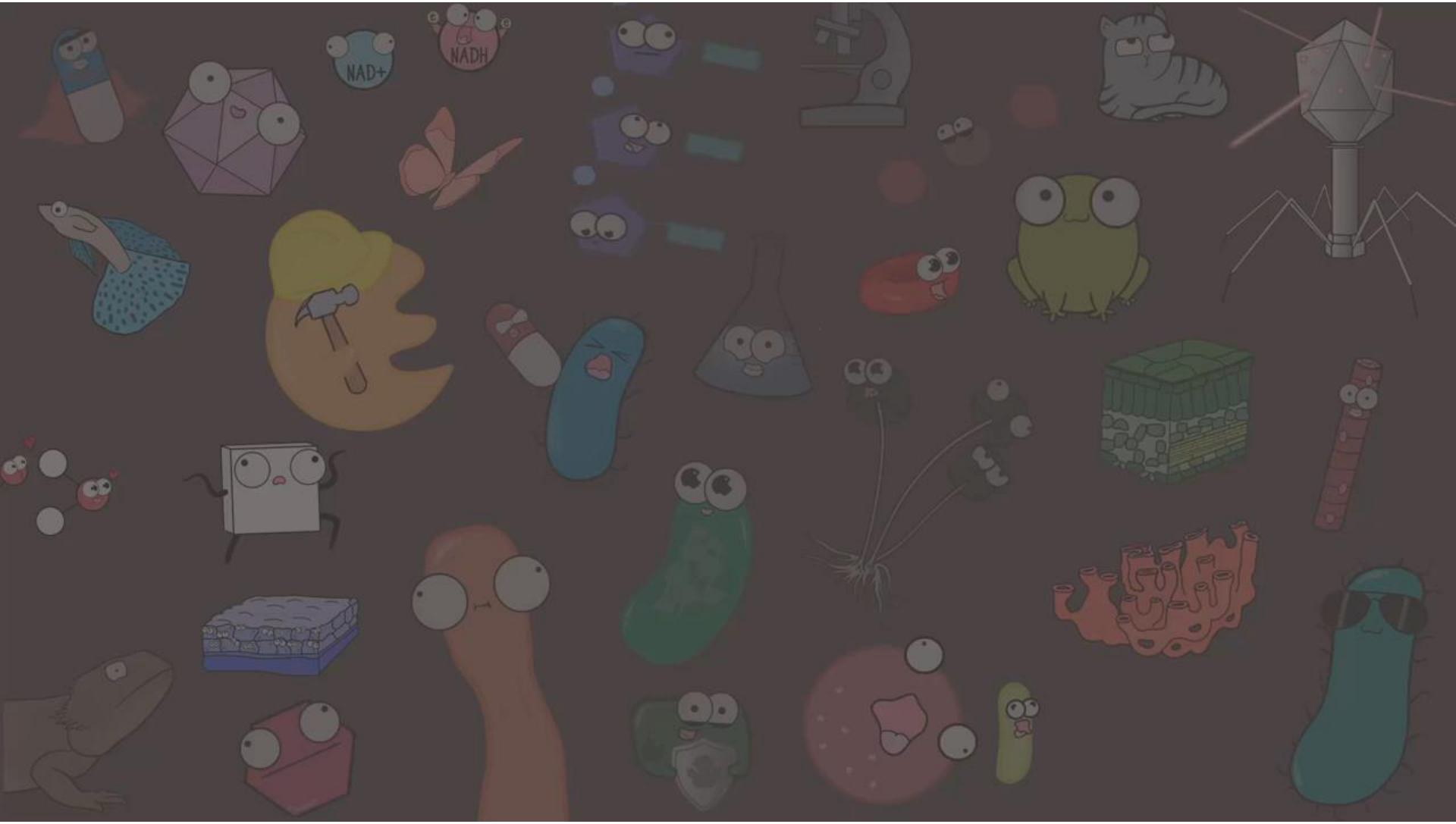

Animals
E. Coli
Thermophiles
Plants
Fungi
Bacteria
Archaea
Eukaryota

| Characteristic | Bacteria | Archaea | Eukarya |
|---|---|---|---|
| Peptidoglycan in cell wall | All | Some | None |
| Ribosome | Prokaryotic | In between | Eukaryotic |
| First aa in protein synthesis | Methionine derivative | Methionine | Methionine |
| Sensitivity to streptomycin and chloramphenicol | Yes | No | No |
| Introns in genes | Very rare | Present in some genes | Present in many genes |
| Histones present | No | Yes | Yes |
| RNA polymerase | One kind | Several kinds | Several kinds |
| Carbon linkage of lipids | Ester | **Ether** | Ester |

| Characteristic | Bacteria | Archaea | Eukarya |
|---|---|---|---|
| Peptidoglycan in cell wall | | | |
| Ribosome | | | |
| First aa in protein synthesis | | | |
| Sensitivity to streptomycin and chloramphenicol | | | |
| Introns in genes | | | genes |
| Histones present | | | |
| RNA polymerase | | | |
| Carbon linkage of lipids | | | |



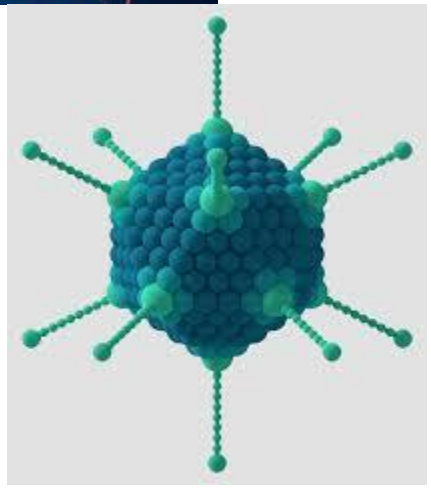
# Archaea

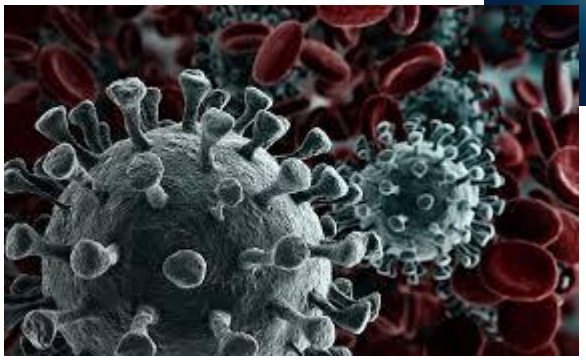


# Part 3. VIRUS



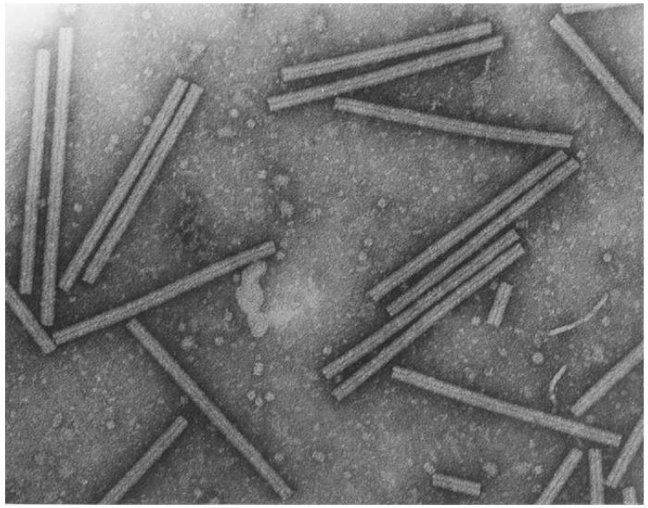






Tobacco mosaic virus (TMV)

Bacteria-free filtrate still infectious





Dimitri Iwanowsky (1864- 1920)

## What are viruses?

Viruses are parasites—depends entirely upon another a host (another living organism )

They cannot be observed using a light microscope

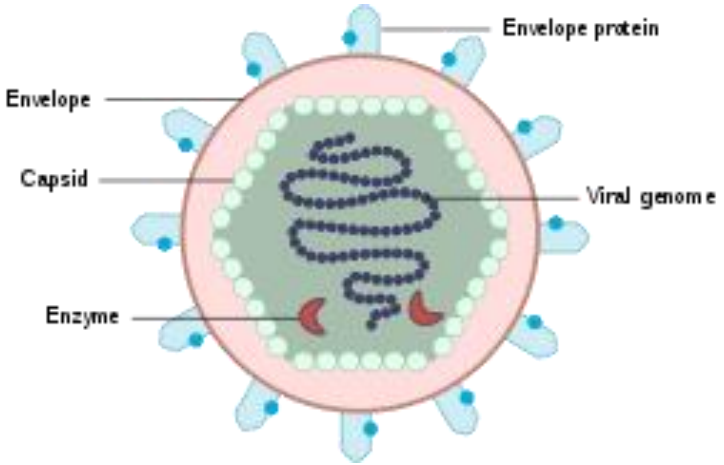
They have no internal cellular structure

They contain either **DNA or RNA, but not both**

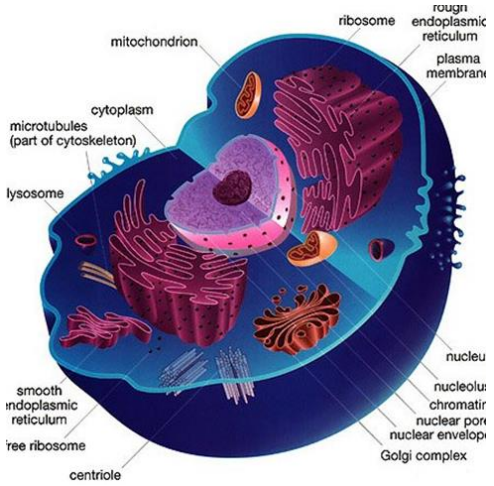
They are incapable of replication unless occupying an appropriate living host cell

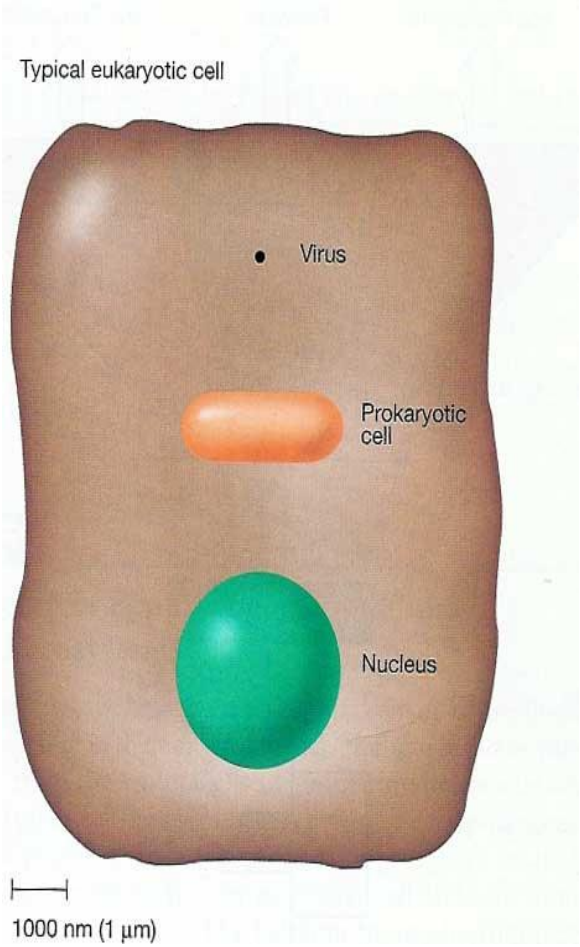
They are incapable of metabolism

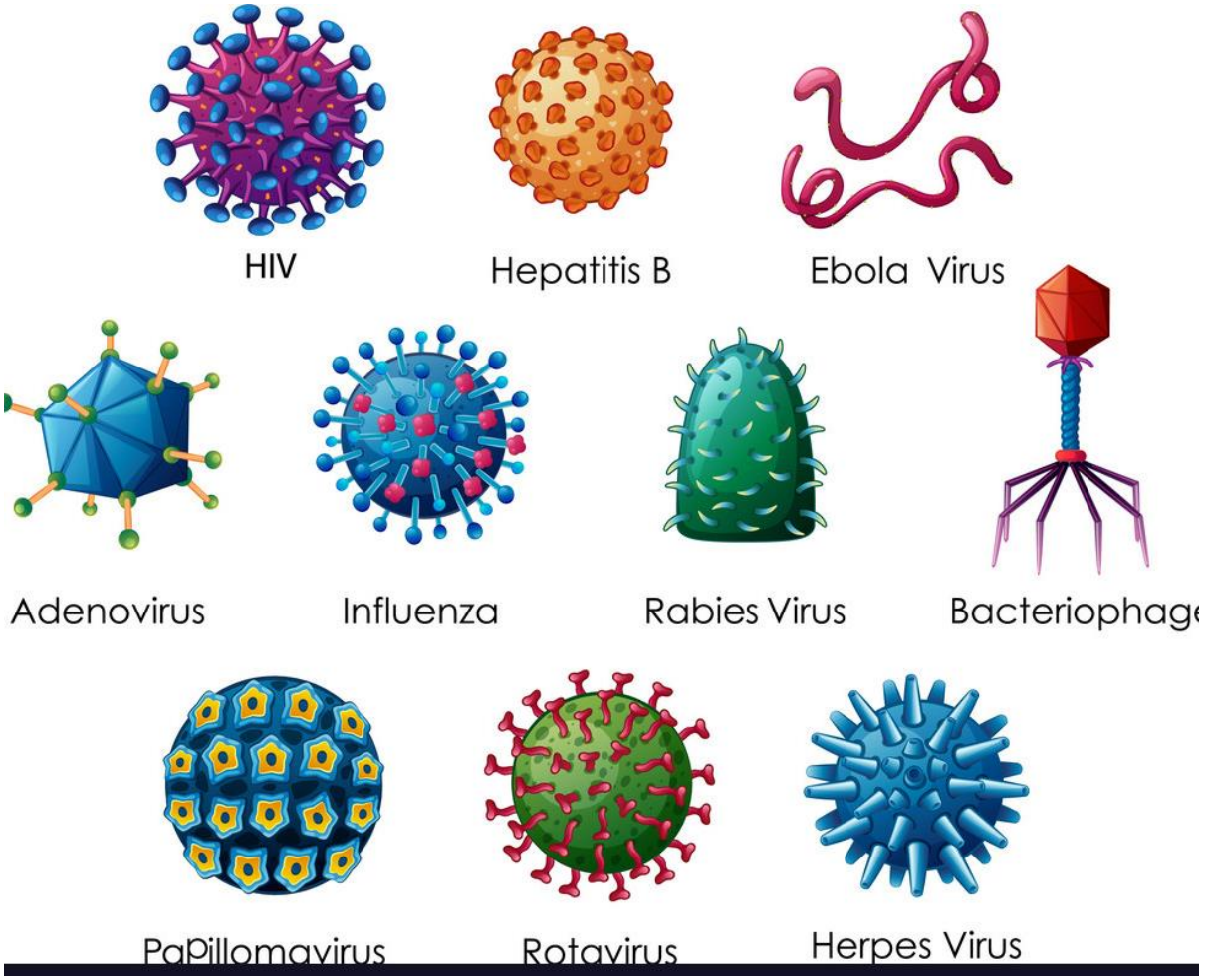
Individuals show no increase in size.



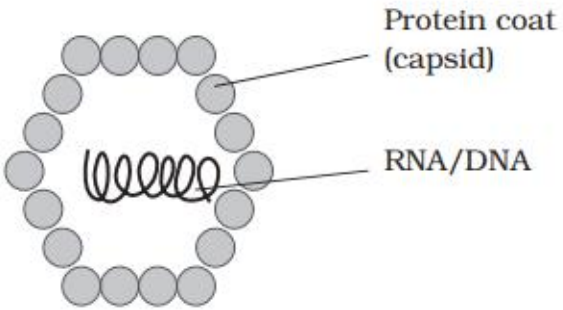

VS



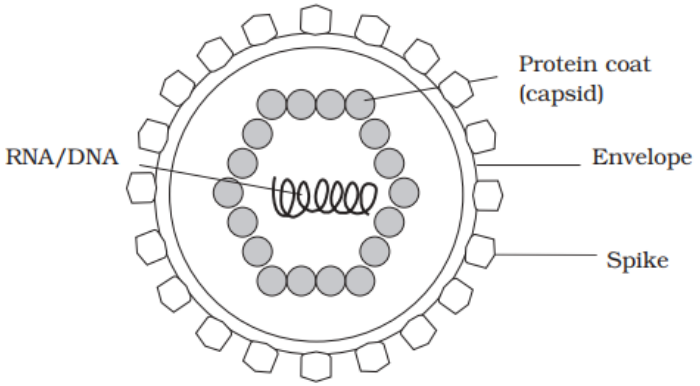

Typical eukaryotic cell
Virus
Prokaryotic cell
Nucleus
1000 nm (1 μm)


HIV
Hepatitis B
Ebola Virus
Adenovirus
Influenza
Rabies Virus
Bacteriophage
Papillomavirus
Rotavirus
Herpes Virus

Viral structure




naked



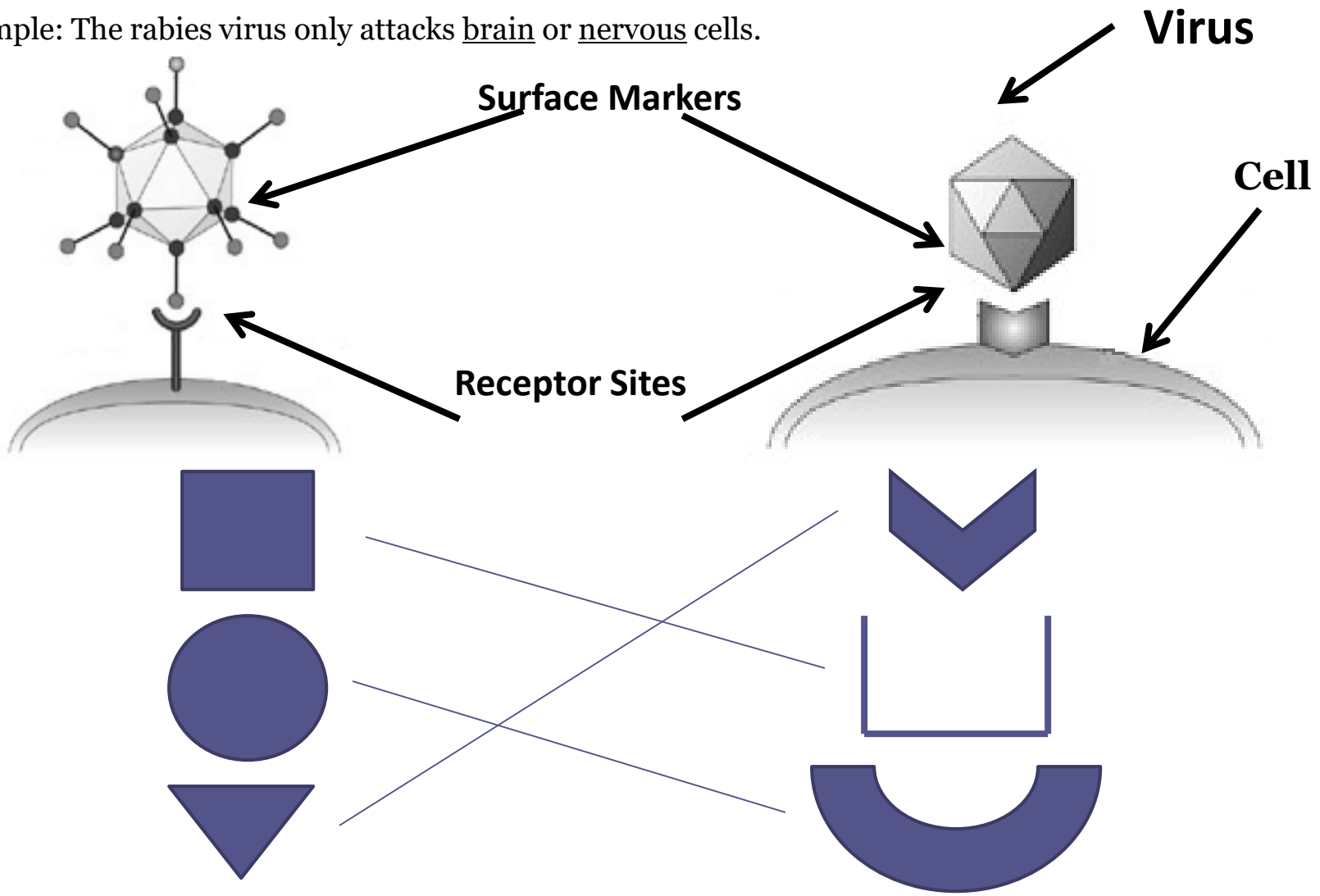

enveloped

Viruses comprise a nucleic acid genome surrounded by a protein coat (capsid)

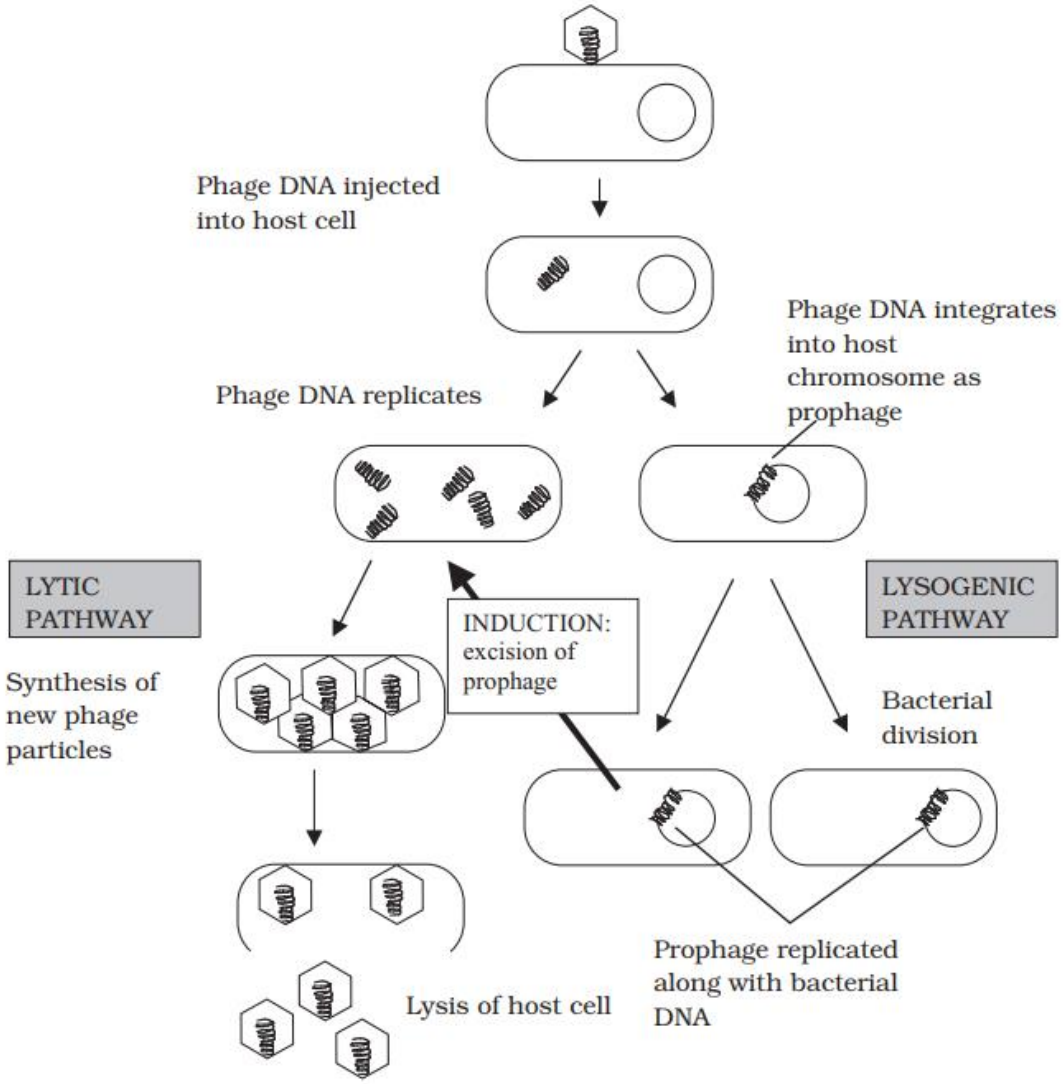
A virus recognizes cells it can infect by matching its **surface marker** with a **receptor site** on a cell.

Certain viruses can only attack certain cell types. They are said to be **specific.**

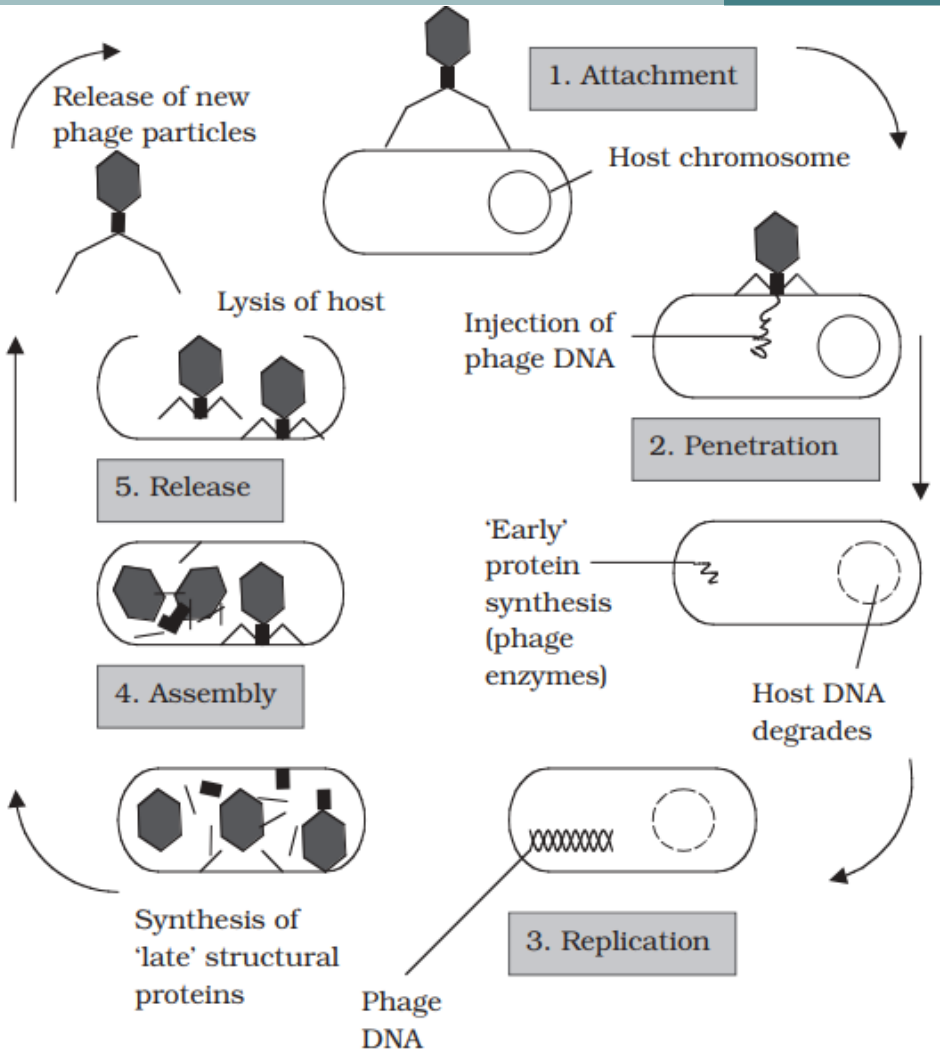
Example: The rabies virus only attacks brain or nervous cells.



# Replication

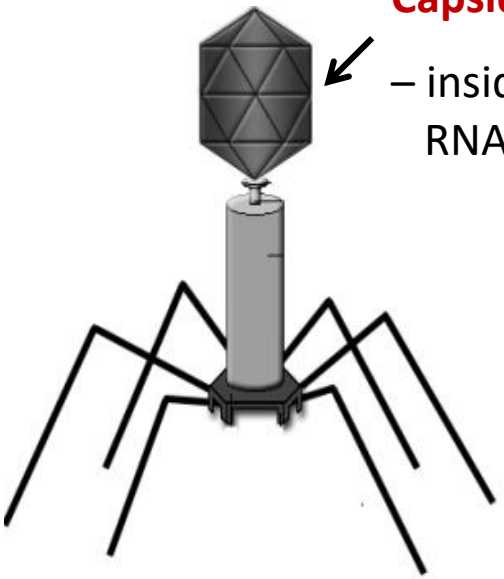



Replication cycle of a temperate phage. In the lysogenic pathway, the phage DNA is integrated as a prophage into the host genome, and replicated along with it. Upon induction by an appropriate stimulus, the phage DNA is removed and enters a lytic cycle



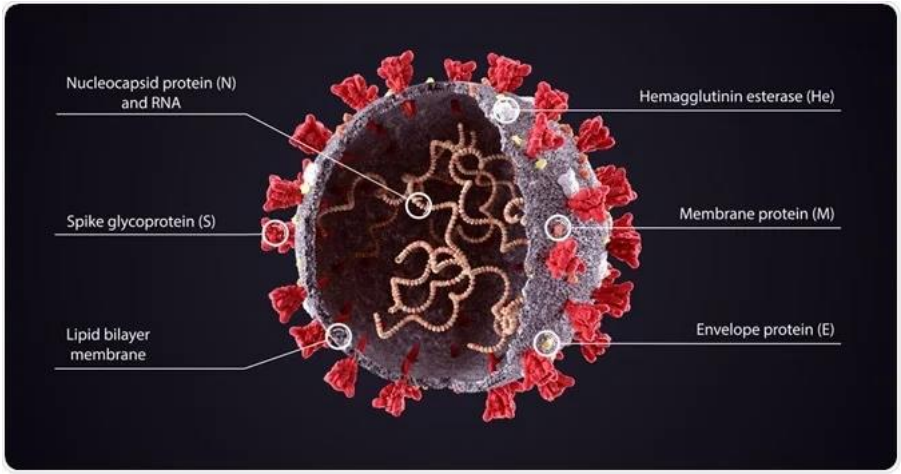

The lytic cycle of phage T4. The cycle comprises the five main stages described in the text; from injection of phage of DNA to cell lysis takes 22 minutes. The number of phage particles released per cell is called the burst size, and for T4 it ranges from 50 to 200

# 1. Bacteriophage—viruses that infect bacteria

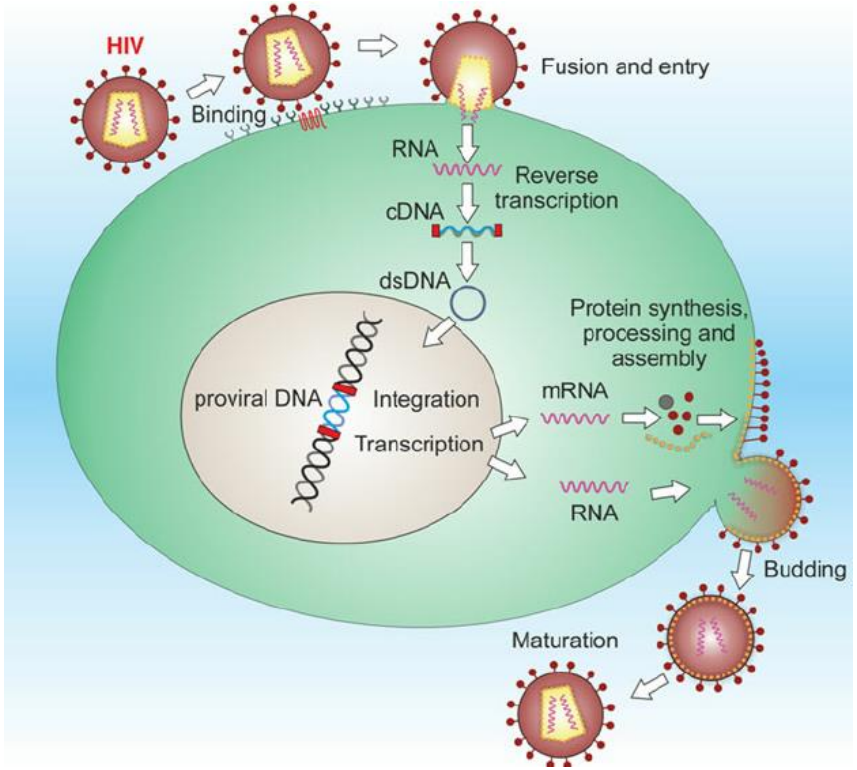


**Capsid** (**protein** coat)

– inside contains either RNA or DNA




## 2. HIV




Viruses with RNA that transcribe into DNA are called retroviruses.

# Prions

A prion (=proteinaceous infectious particle) is a self-replicating protein responsible for a range of neurodegenerative disorders in humans and mammals.

These include bovine spongiform encephalopathy (BSE, ‘mad cow disease’) and its human equivalent, Creutzfeldt–Jakob disease.

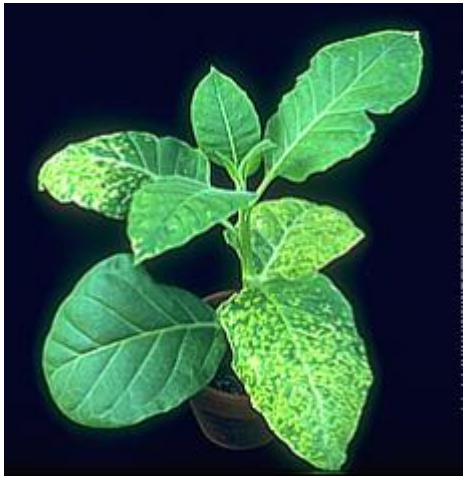
Prions (affects the brain and is always fatal)



# Viroids

A viroid is a plant pathogen that comprises only ssRNA and does not code for a protein product

In 1971, Theodor Diener proposed the name viroid to describe a newly discovered pathogen of potatoes.
Viroids are many times smaller than the smallest virus, and consist solely of a small circle of ssRNA containing some 300–400 nucleotide bases and no protein coat



| | Virus | Cells |
|---|---|---|
| *Structure* | **RNA or DNA core (center), protein coat (capsid)** | **Cell membrane, cytoplasm, genetic material, organelles** |
| *Reproduction* | **Copies itself only inside host cell-- REPLICATION** | **Asexual or Sexual** |
| *Genetic Material* | **DNA or RNA** | **DNA and RNA** |
| *Growth and Development* | **NO** | **YES—Multicellular Organisms** |
| *Obta… Ener…* | | |
| *Resp… Env…* | …NO | |
| *Change over time* | **NO** | **YES** |

How many characteristics of life do viruses possess?

# ARE VIRUSES LIVING?

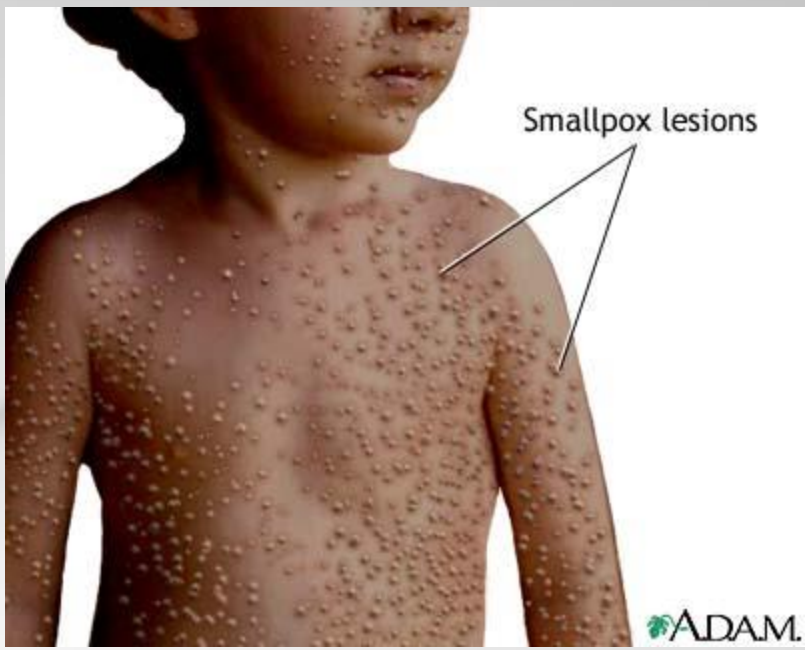


Importance:

*Harmful

Causes disease—pathogenic

Disease producing agent—pathogen




Human Diseases: Warts, common cold, Influenza (flu), Smallpox, Ebola, Herpes, AIDS, Chicken pox, Rabies

Viruses disrupt the body's normal equilibrium/balance

Viruses can be prevented with vaccines, but NOT treated with antibiotics.

(antibiotics treat bacteria)




Beneficial:

Genetic Engineering—harmless virus carries good genes into cells.

